

Computer Networks (01)

An Introduction

lijt@fudan.edu.cn

Some slides from Jim Kurose

Course Non-goals

❑ Learn how to configure a Router

- ◆ That requires a class all by itself
- ◆ Huawei/Cisco teaches those classes
- ◆ Our perspective will be broader

❑ Become "Internet Experts"

- ◆ The Internet will be our frequent *motivating example*
- ◆ *Our perspective will be broader*
- ◆ *Focus on Internet, but understand important common issues and challenges*

Why *not* an "Internet class"?

- **Is there anything other than the Internet?**
Give two examples ?

Course Goals

- ❑ Think “the network way”
 - ◆ Distributed coordination is hard
- ❑ Learn how computer networks work
 - ◆ Problems, approaches, protocols, software
- ❑ Learn how to write network applications
- ❑ Hands-on understanding of network internals
 - ◆ Course Projects

Chapter 1: introduction

Overview/roadmap:

- What *is* the Internet? What *is* a protocol?
- **Network edge:** hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- **Performance:** loss, delay, throughput
- Protocol layers, service models
- Security
- History

What Is a Network?

- ❑ Collection of nodes and links that connect them
- ❑ This is vague. Why? Consider different networks:
 - ❖ Internet
 - ❖ Social networks
 - ❖ Telephone
 - ❖ Others
 - - transport nets, sensor nets, cellular nets, ...

What is a Computer Network

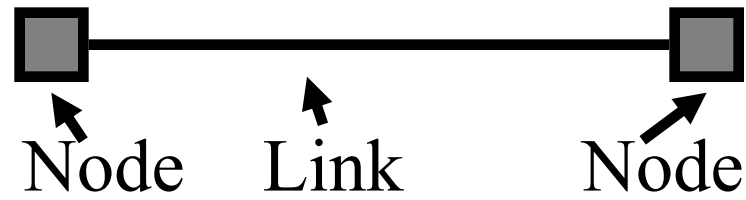
- ❑ Interconnected;
- ❑ Autonomous;
- ❑ Collection.

Abstraction:

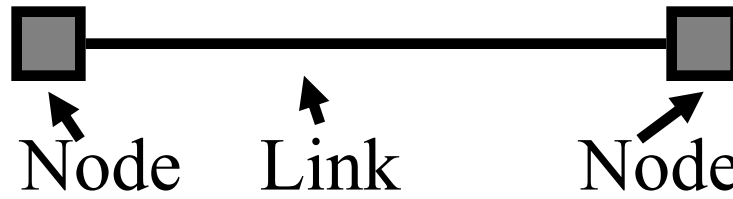
End system + Link + Switching node

Host + Link + Router (for Internet)

How to Draw a Computer Network

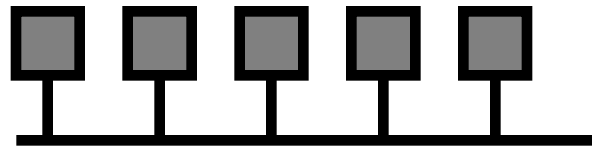


Building block: The Links

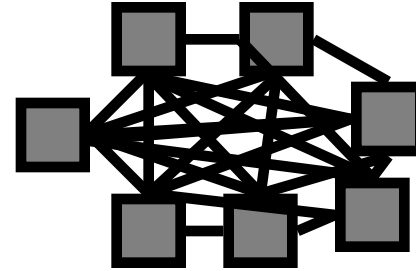


- ❑ Electrical questions
 - ❖ Voltage, frequency, ...
 - ❖ Wired or wireless?
- ❑ Link-layer issues: How to send data?
 - ❖ When to talk - can everyone talk at once?
 - ❖ What to say - low-level format?
 - ❖ Stay tuned for Chapter 6
- ❑ Okay... what about more nodes?

□ ... But what if we want more hosts?



One wire

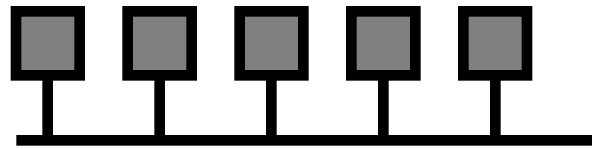


Wires for everybody!

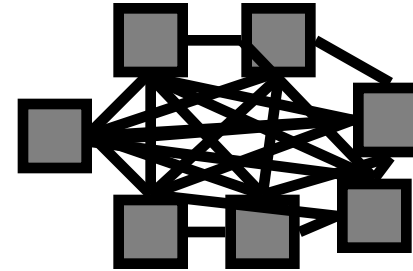
□ Scalability?!

Scalability: how to solve ?

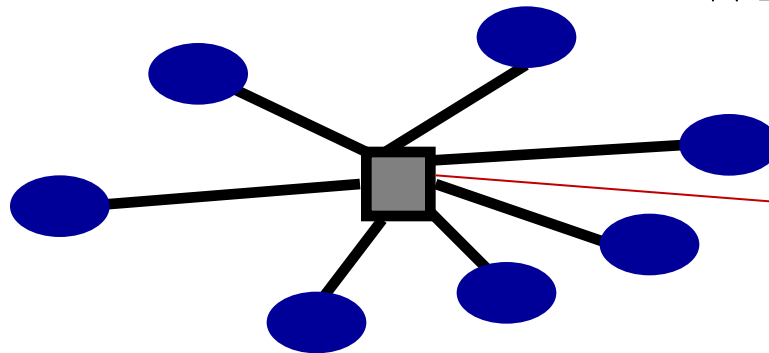
□ ... But what if we want more hosts?



One wire



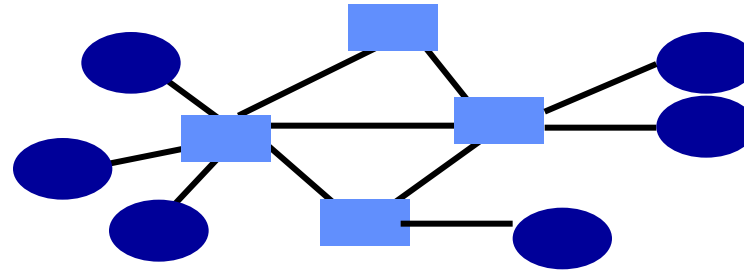
Wires for everybody!



Switching node

Multiplexing!

- ❑ Need to share network resources



- ❑ How? Switched network
 - ❖ Party "A" gets resources sometimes
 - ❖ Party "B" gets them sometimes
- ❑ Interior nodes act as "Routers" or "Switches"
- ❑ What mechanisms can share resources?

Review:

□ Layering

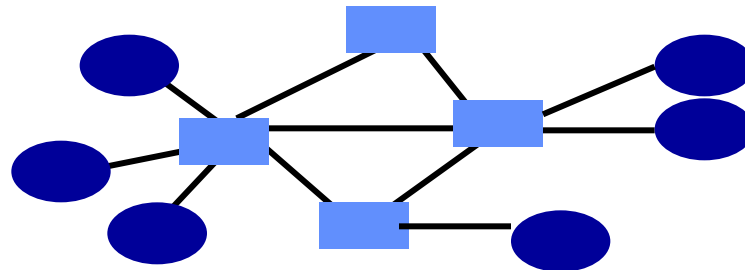
- ❖ Abstraction
- ❖ OSI vs. TCP/IP

□ Control plane and Data Plane

- ❖ Design considerations

□ Basic Model for computer networks

End system + Link + Switching node

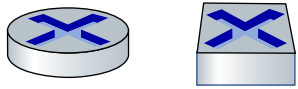


The Internet: a "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers, switches*

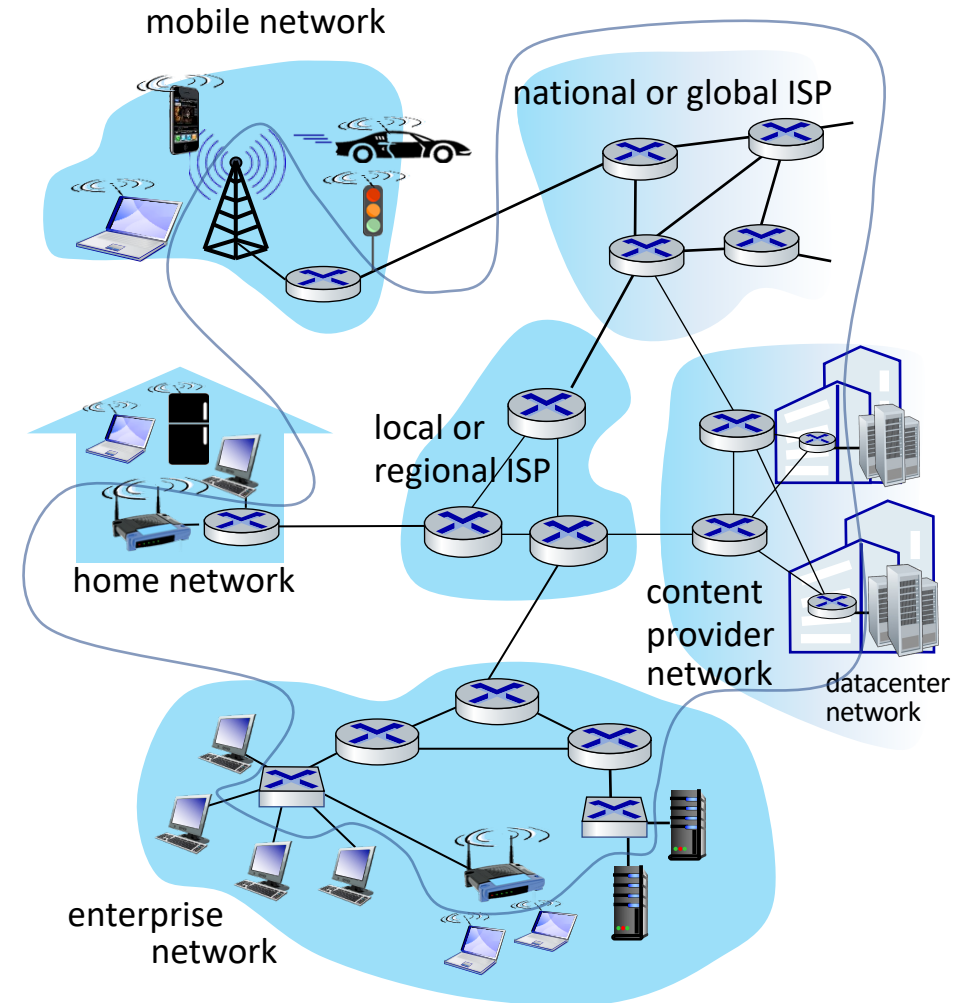
Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*



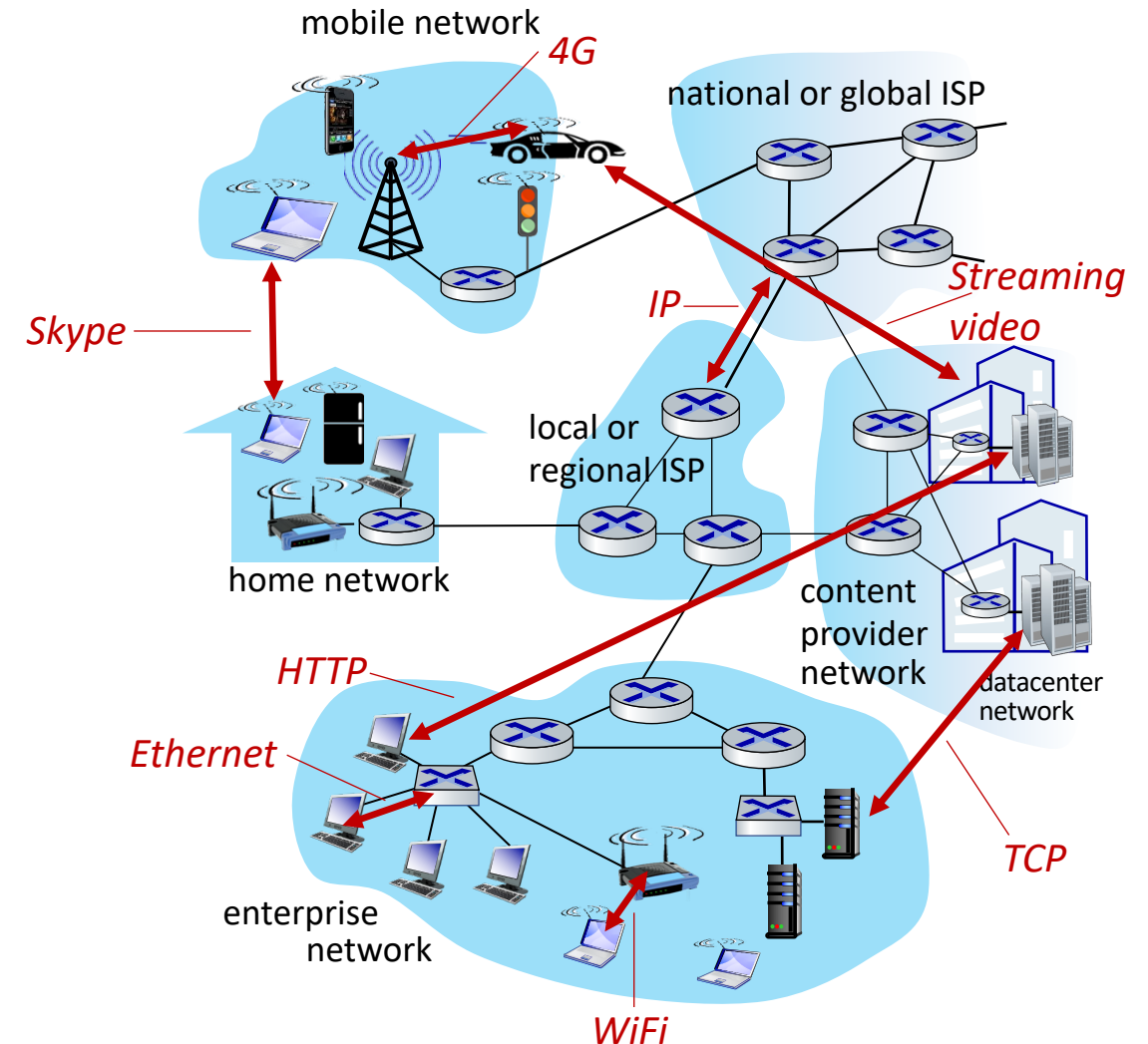
Networks

- collection of devices, routers, links: managed by an organization



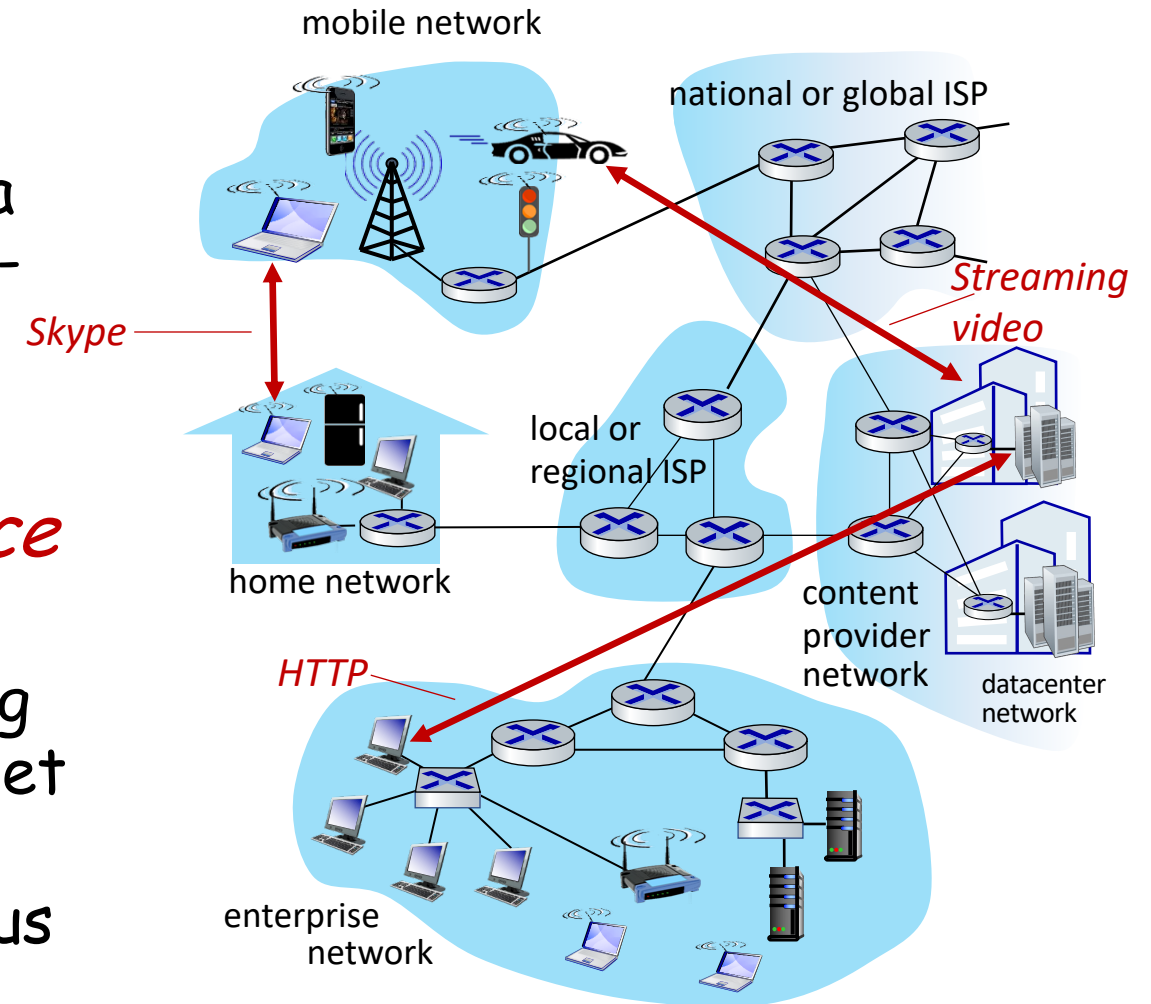
The Internet: a "nuts and bolts" view

- *Internet: "network of networks"*
 - Interconnected ISPs
- *protocols are everywhere*
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a "services" view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
 - provides service options, analogous to postal service



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

- ... specific messages sent
- ... specific actions taken
when message received,
or other events

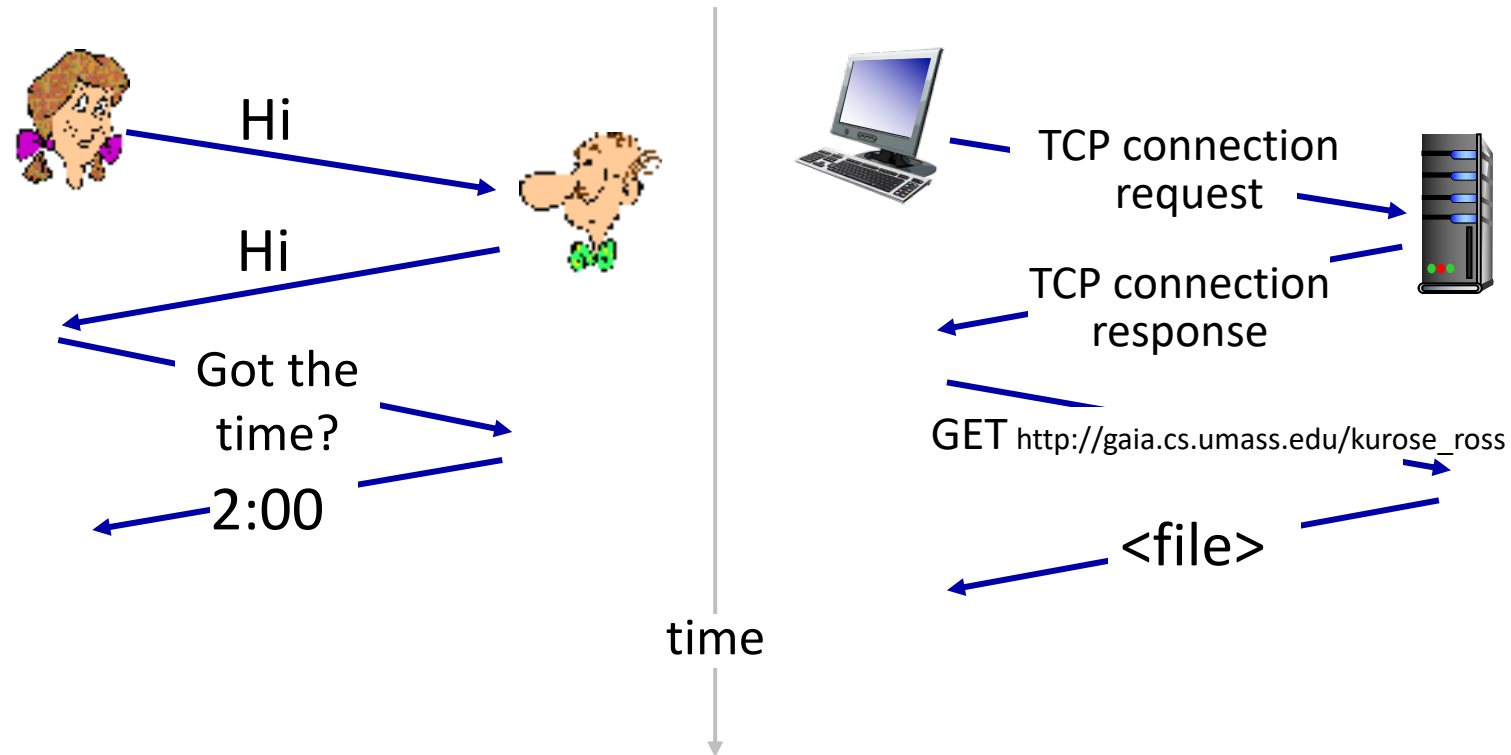
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on message transmission, receipt*

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

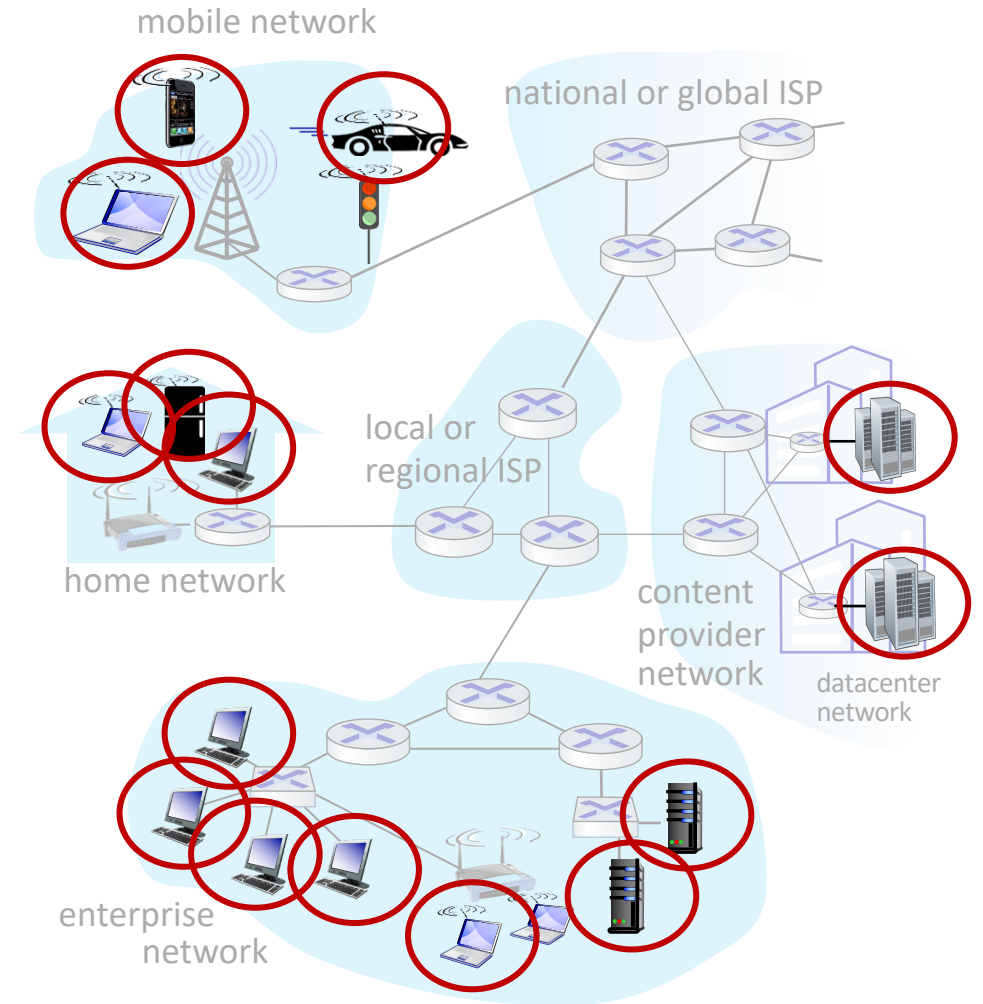
Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- **Network edge:** hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



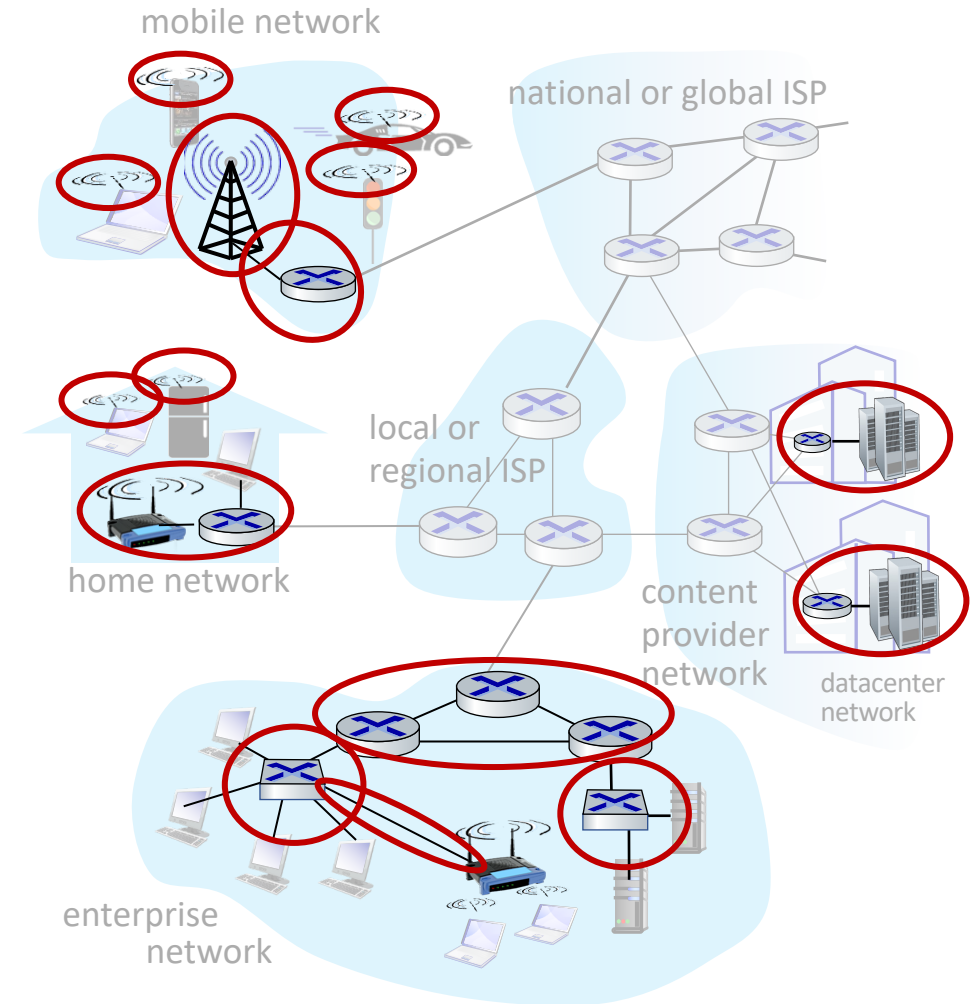
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

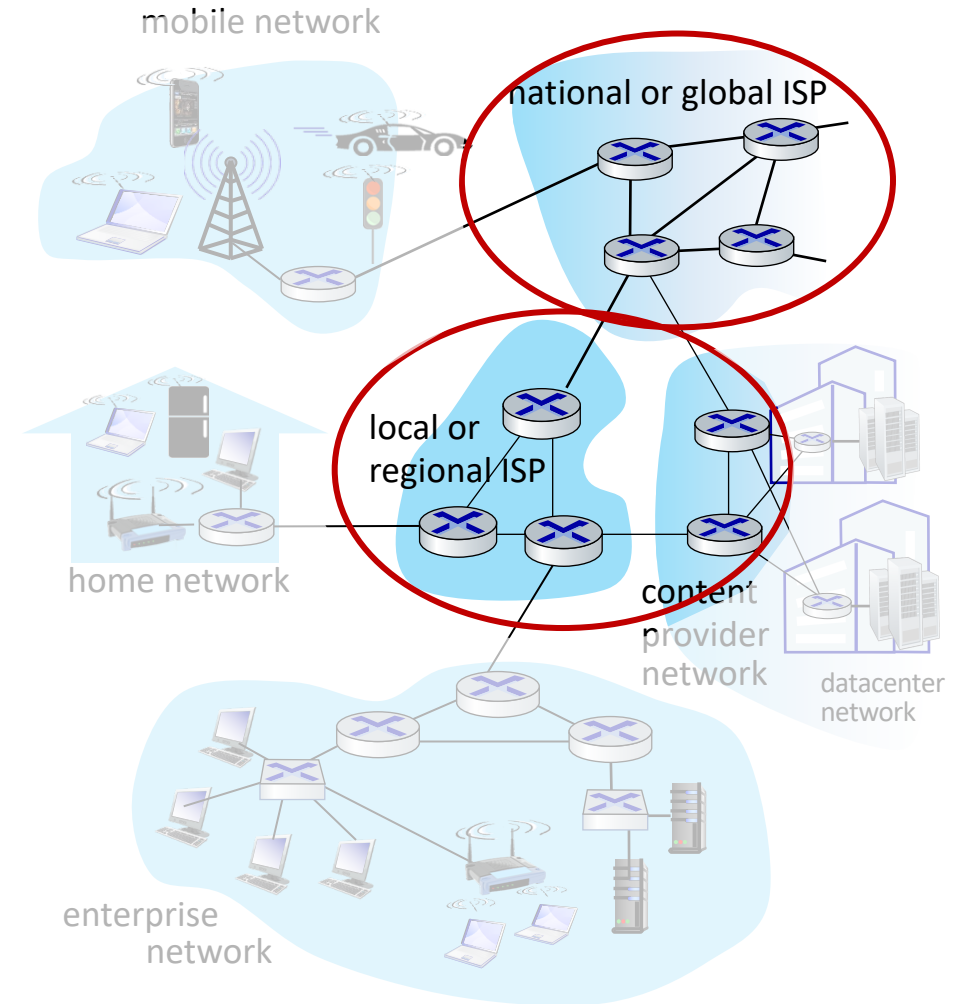
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links

Network core:

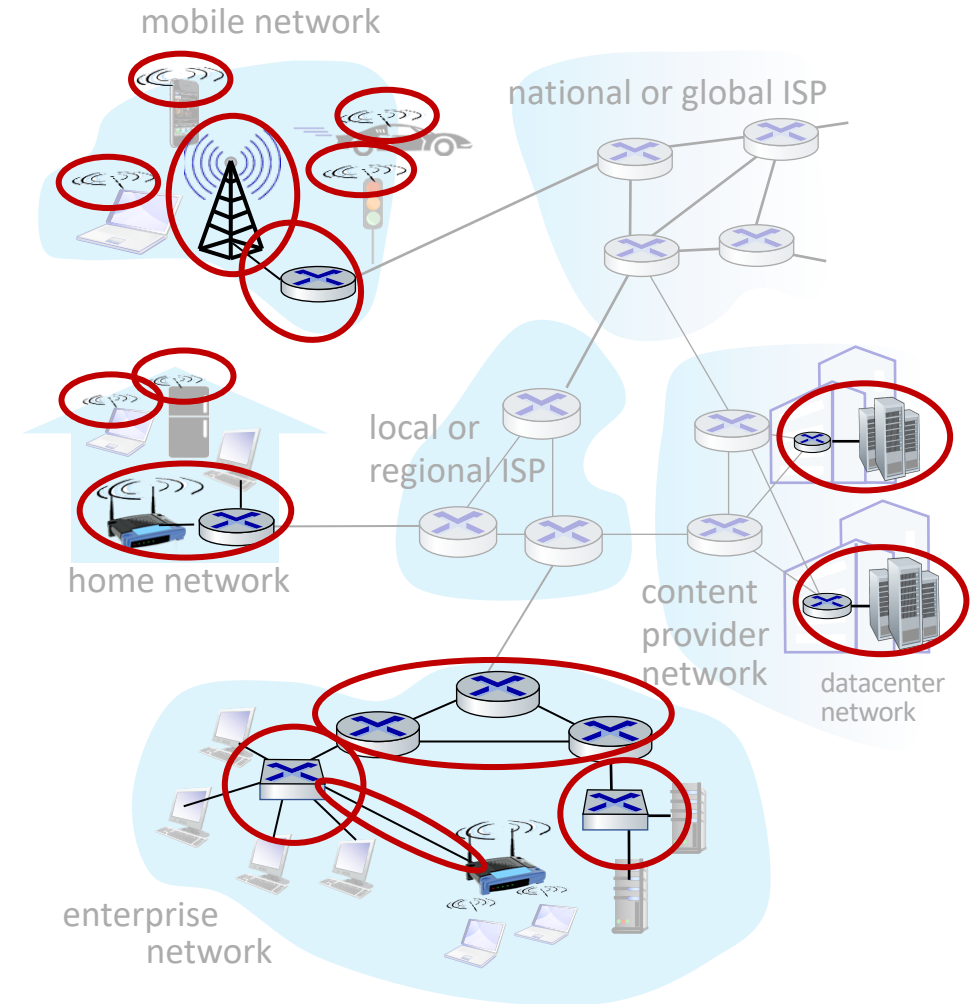
- interconnected routers
- network of networks



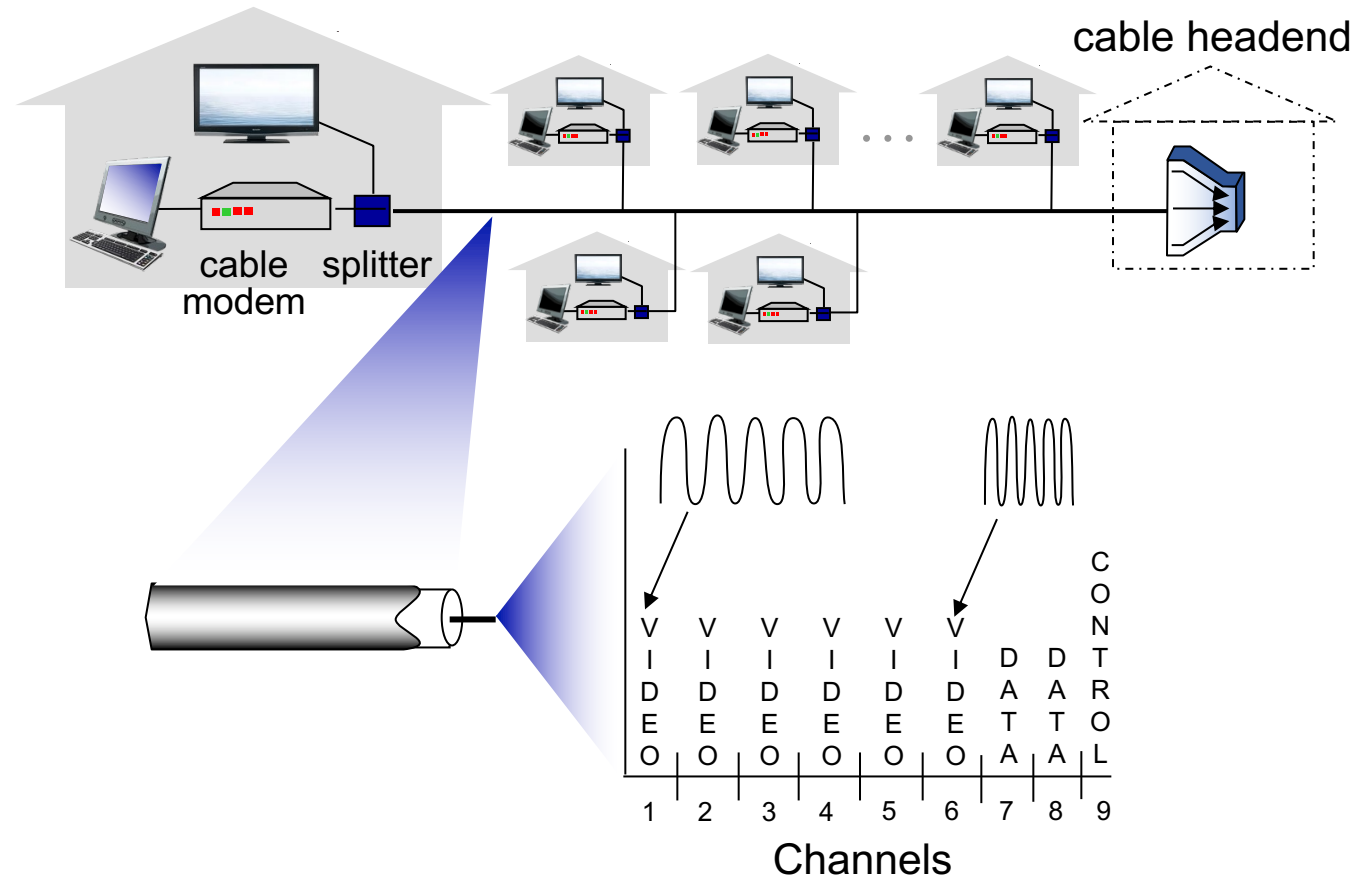
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

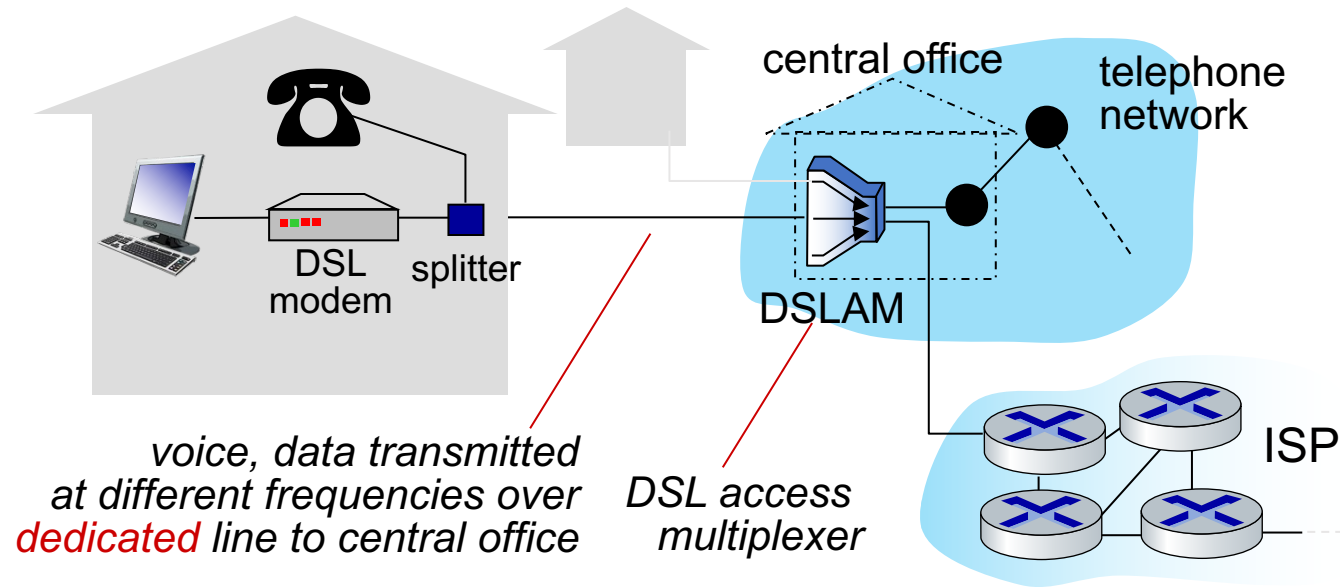


Access networks: cable-based access



frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access networks: digital subscriber line (DSL)

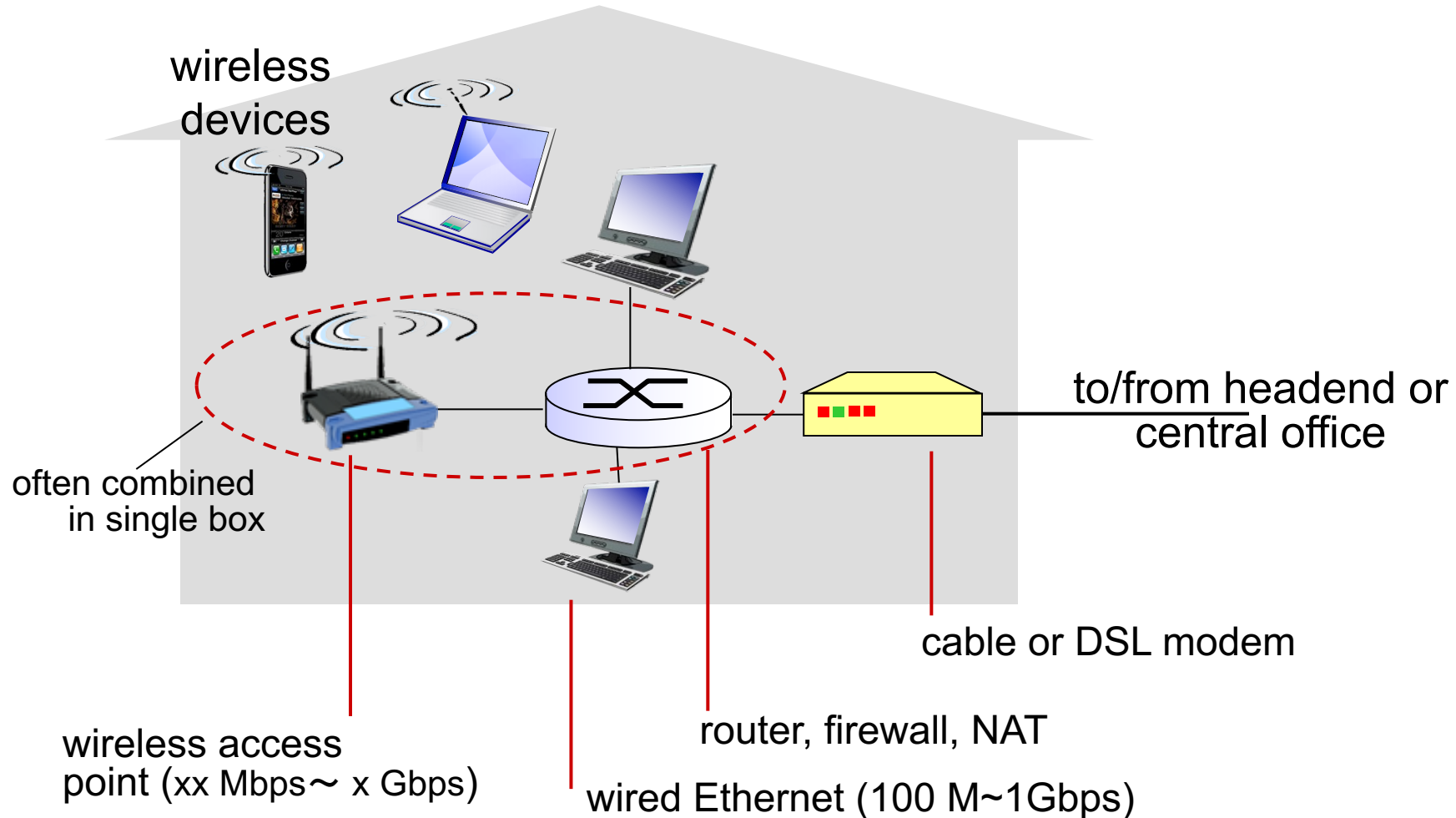


- use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access: Home networks

Typical home network components:

- ADSL, cable modem, or FTTH (eg. EPON)
- router/firewall/NAT
- Ethernet
- wireless access point



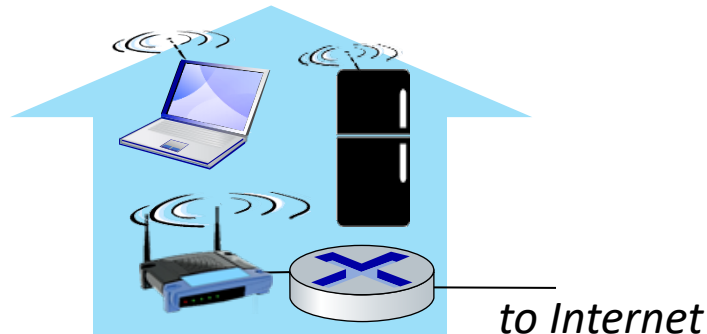
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

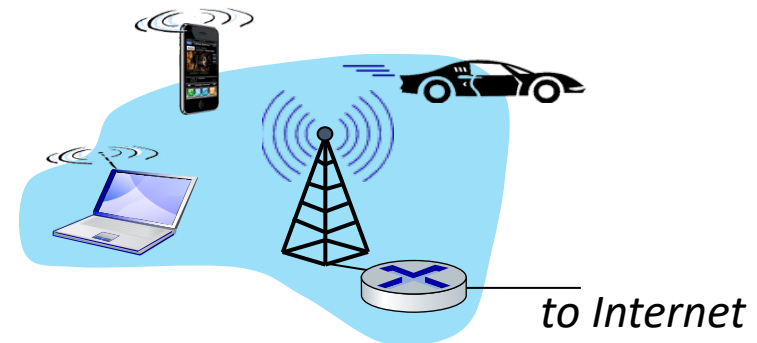
Wireless local area networks (WLANs)

- within building (50 m)
- 802.11b/g/n (WiFi): 11, 54, 450Mbps, 9.6Gbps(WiFi 6) transmission rate

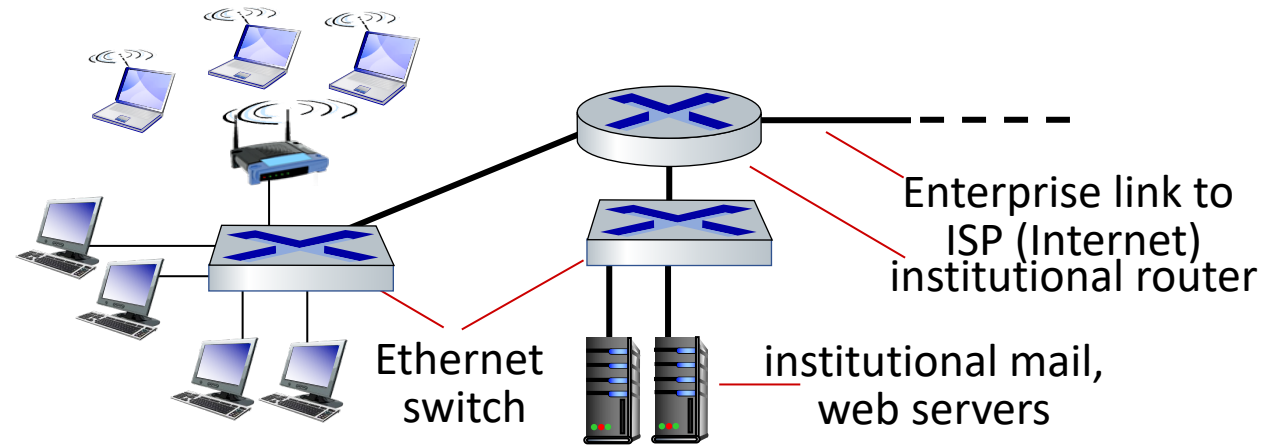


Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps, 100Mbps ~ x Gbps
- 4G/5G cellular networks



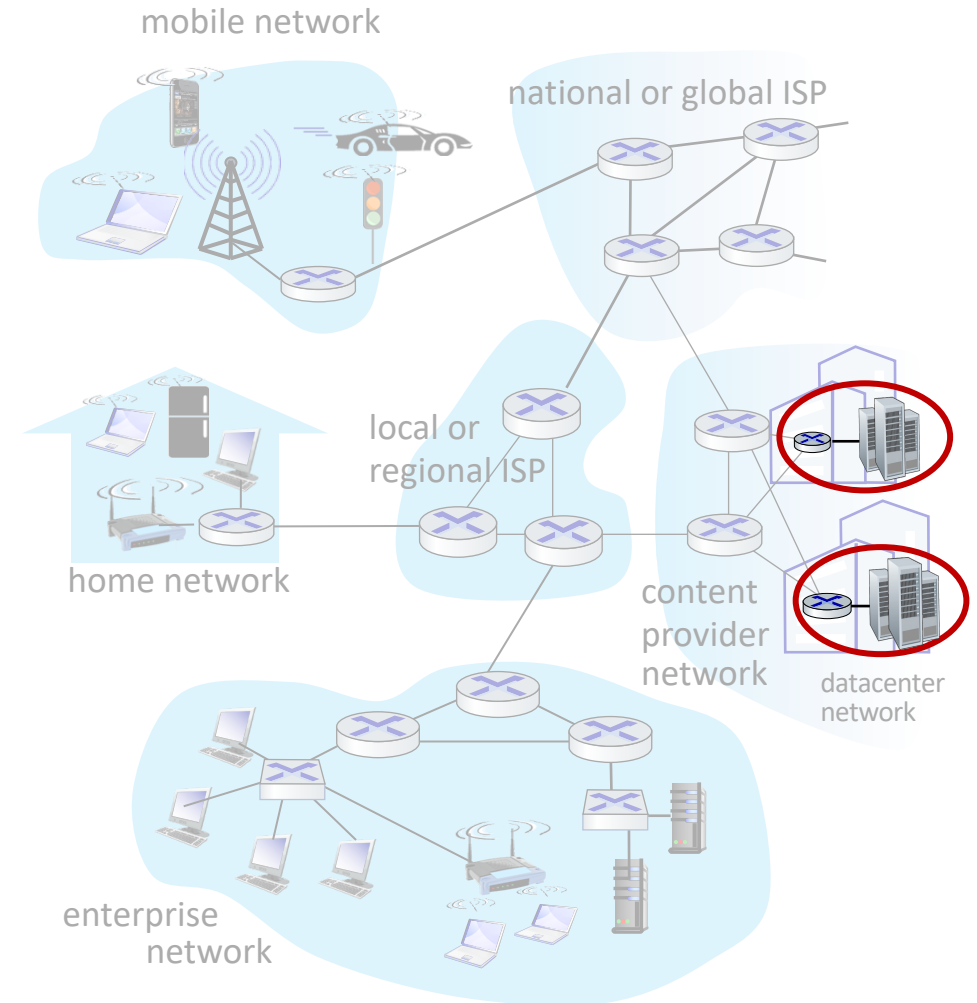
Access networks: enterprise networks



- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps ...

Access networks: data center networks

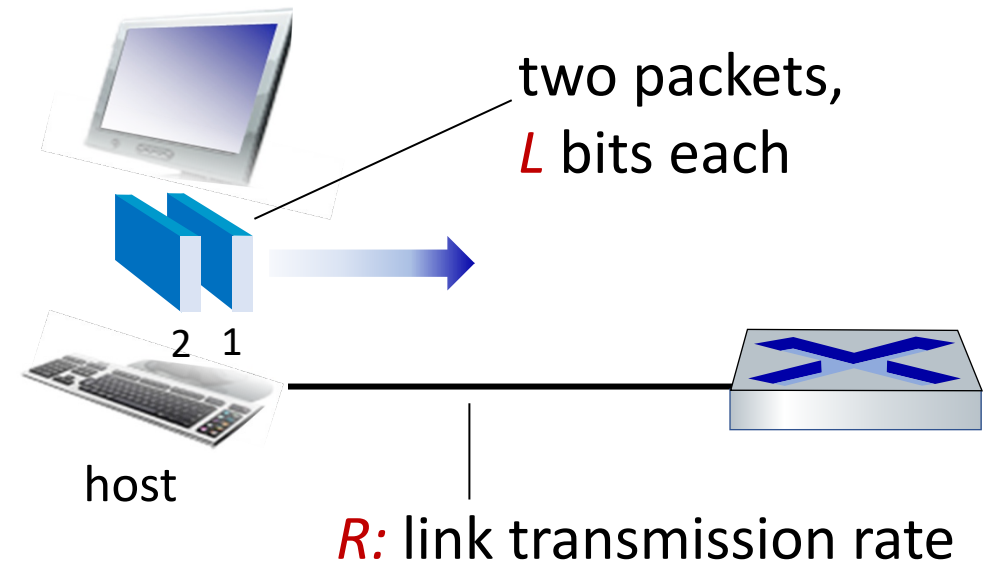
- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Host: sends *packets* of data

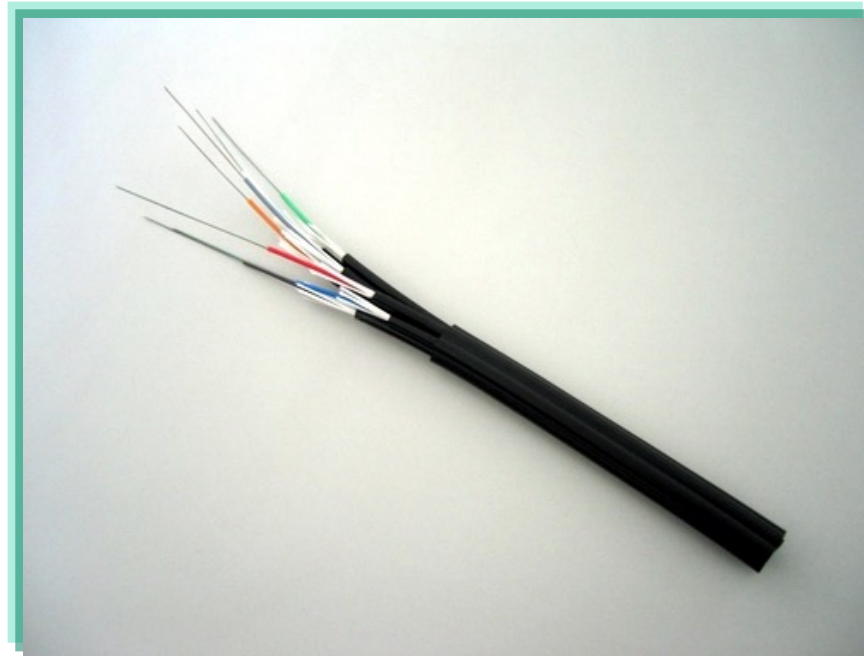
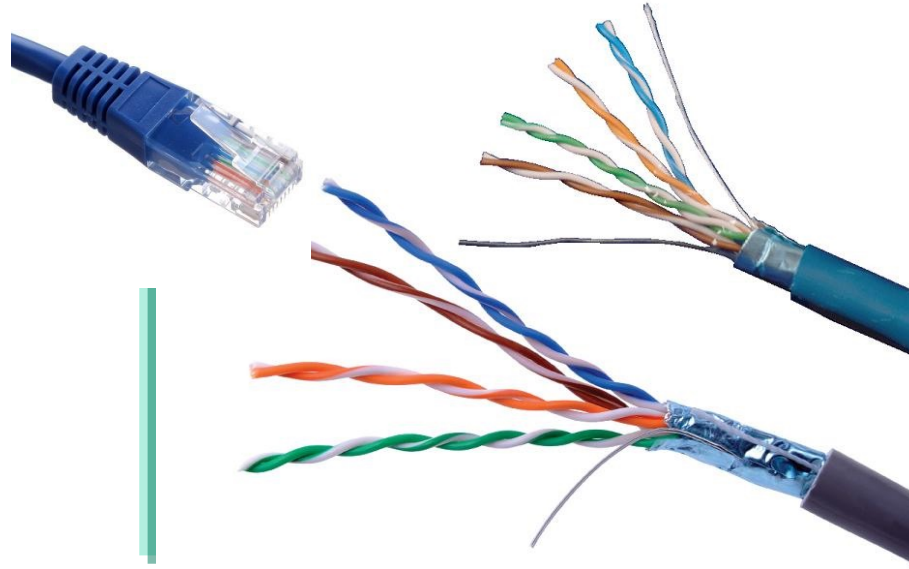
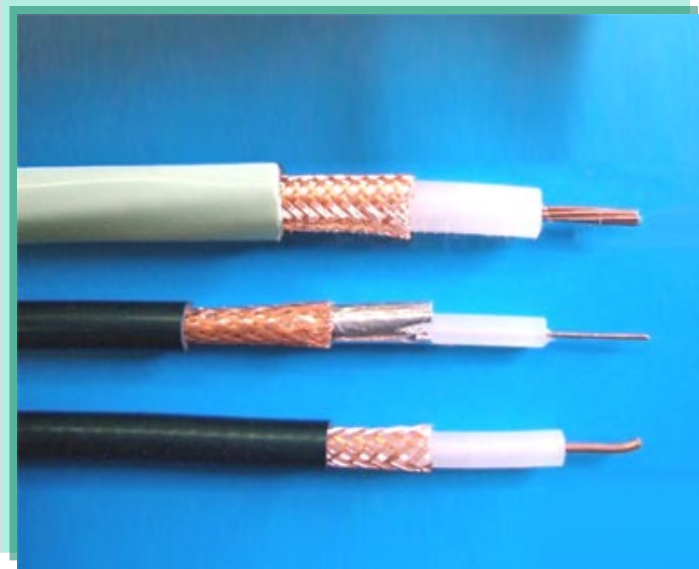
host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate* R
 - link transmission rate, aka link *capacity, aka link bandwidth*



$$\begin{array}{l} \text{packet} \\ \text{transmission} \\ \text{delay} \end{array} = \begin{array}{l} \text{time needed to} \\ \text{transmit } L\text{-bit} \\ \text{packet into link} \end{array} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Physical Media

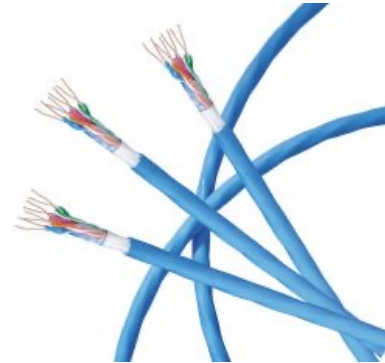


Links: physical media

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Links: physical media

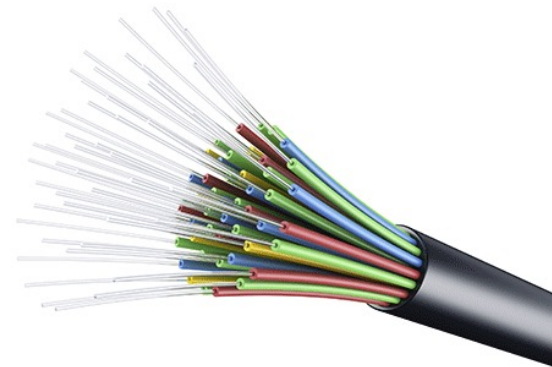
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: physical media

Wireless radio

- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

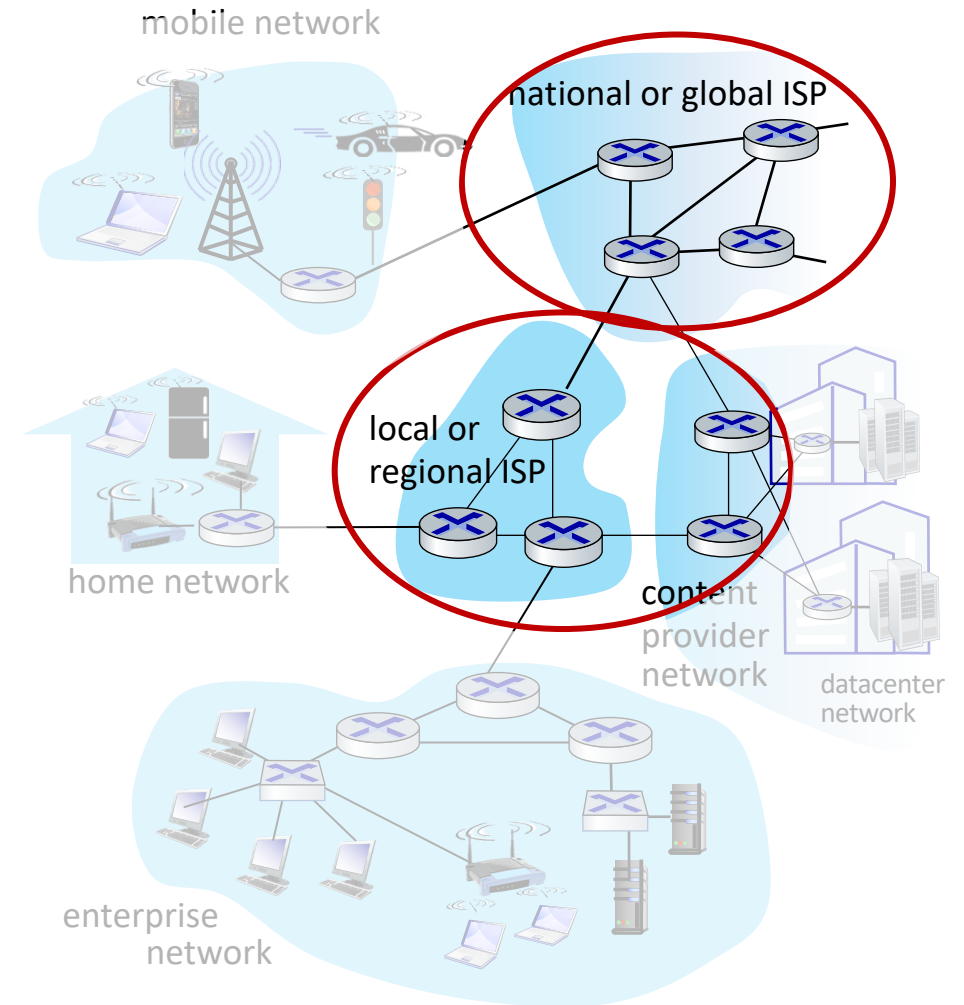
- **Wireless LAN (WiFi)**
 - 10-100's Mbps; 10's of meters
- **wide-area** (e.g., 4G/5G cellular)
 - over ~10 Km
 - 3G: ~2Mbps; 4G: 1xxMbps
 - 5G: [down]: 1Gbps, 10G(in theory); [up]: 100Mbps
- **Bluetooth**: cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History

The network core

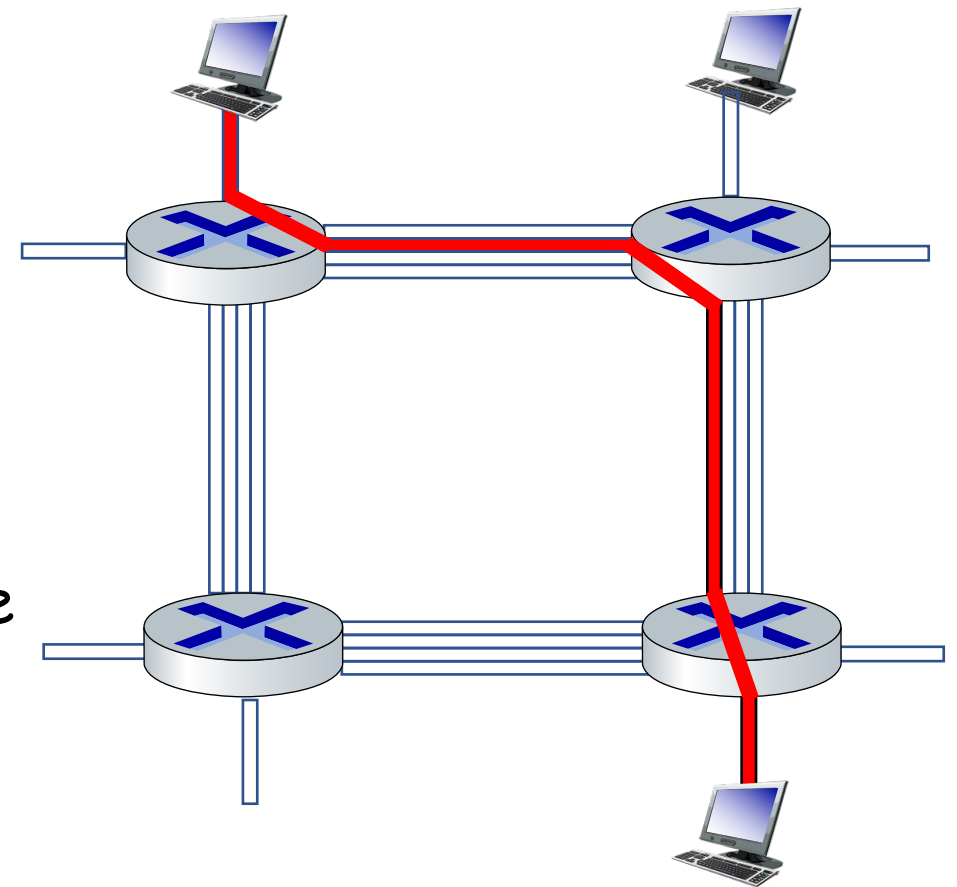
- mesh of interconnected switches/routers
- **packet-switching**: hosts break application-layer messages into *packets*
 - network **forwards** packets from one switches /router to the next, across links on path from **source to destination**



Network core: Circuit switching

end-end resources allocated to,
reserved for "call" between source
and destination

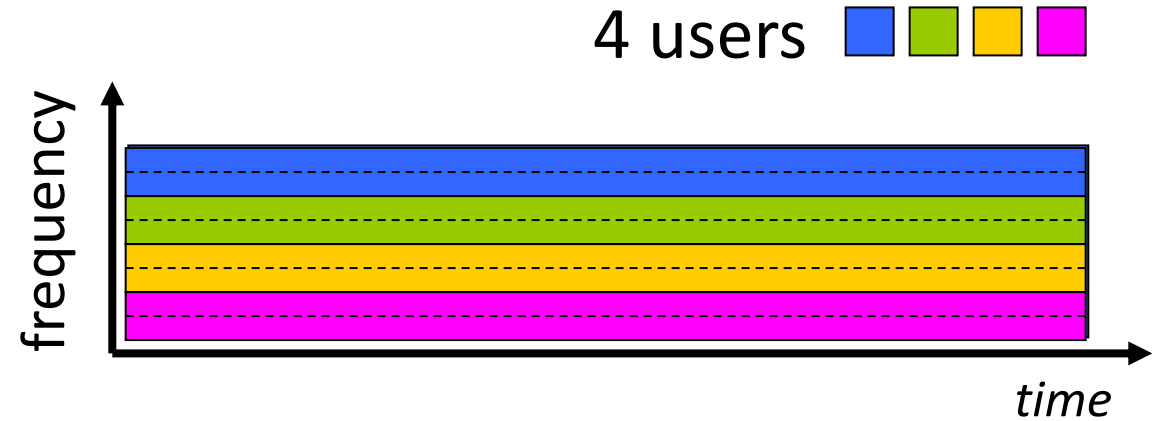
- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- circuit segment idle if not used by call
(no sharing)
- commonly used in traditional telephone networks



Circuit switching: FDM and TDM

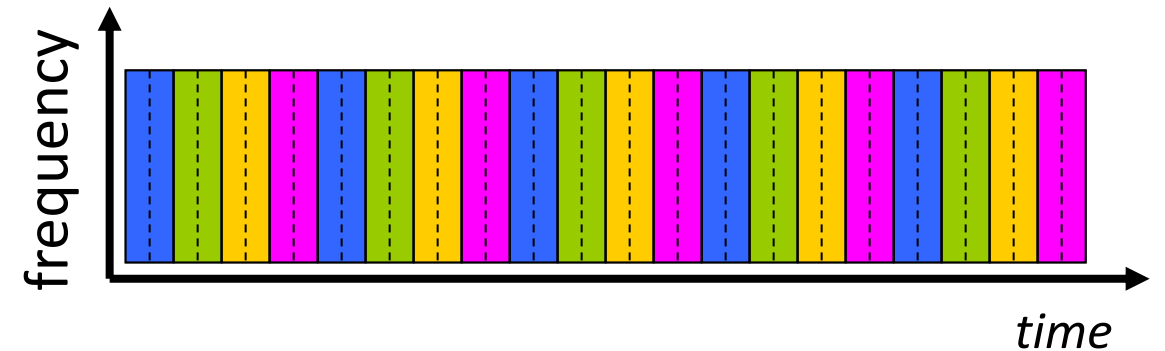
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



More Examples

GSM (a single cell)

- 4 Carriers --FDM
- 8 time slots for each carrier --TDM

	0	1	2	3	4	5	6	7
TRX 1	CCH*	SDCCH/8	TCH	TCH	TCH	TCH	TCH	TCH
TRX 2	TCH	TCH	TCH	TCH	TCH	TCH	TCH	TCH
TRX 3	TCH	TCH	TCH	TCH	TCH	TCH	TCH	TCH
TRX 4	TCH	TCH	TCH	TCH	TCH	TCH	TCH	TCH

T1/DS1 (Digital Signal Designator)

- ❑ In the T1 system, voice or other analog signals are sampled 8,000 times a second and each sample is digitized into an 8-bit word.
- ❑ With 24 channels being digitized at the same time, a 192-bit/slot frame (24 channels each with an 8-bit word) is thus being transmitted 8,000 times a second.
- ❑ Each frame is separated from the next by a single bit, making a 193-bit block. The 192 bit frame multiplied by 8,000 and the additional 8,000 framing bits make up the T1's 1.544 Mbps data rate.

T1 Frame Format



- Each DS0 called a **time slot**
- $8000 \text{ frames/sec} * 8 \text{ bits/slot} = 64 \text{ Kbps}$
- $24 * 8 + 1 = 193 \text{ bits/frame}$
- $8000 \text{ frames/sec} * 193 \text{ bits/frame} = 1.544 \text{ Mbps}$
- 8000 Framing bits sent per second

Transmission Rates

Digital Signal Designator	Data Rate	DS0 Multiple	T-Carrier
DS0	64 Kbps	1	-
DS1	1.544 Mbps	24	T1
DS3	44.736 Mbps	672	T3
OC3	155.52 Mbps		Optical Carrier
OC12	622.08 Mbps		
OC192	10 Gbps		
OC256	13.271 Gbps		
OC768	40 Gbps		

Numerical example

- How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
 - ❖ All links are 1.536 Mbps
 - ❖ Each link uses TDM with 24 slots/sec
 - ❖ 500 msec to establish end-to-end circuit

Let's work it out!

Numerical example

- How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
 - ❖ All links are 1.536 Mbps
 - ❖ Each link uses TDM with 24 slots/sec
 - ❖ 500 msec to establish end-to-end circuit

Let's work it out! (10.5sec)

Network Core: Packet Switching

each end-end data stream
divided into *packets*

- ❑ user A, B packets *share* network resources
- ❑ each packet uses full link bandwidth
- ❑ resources used *as needed*

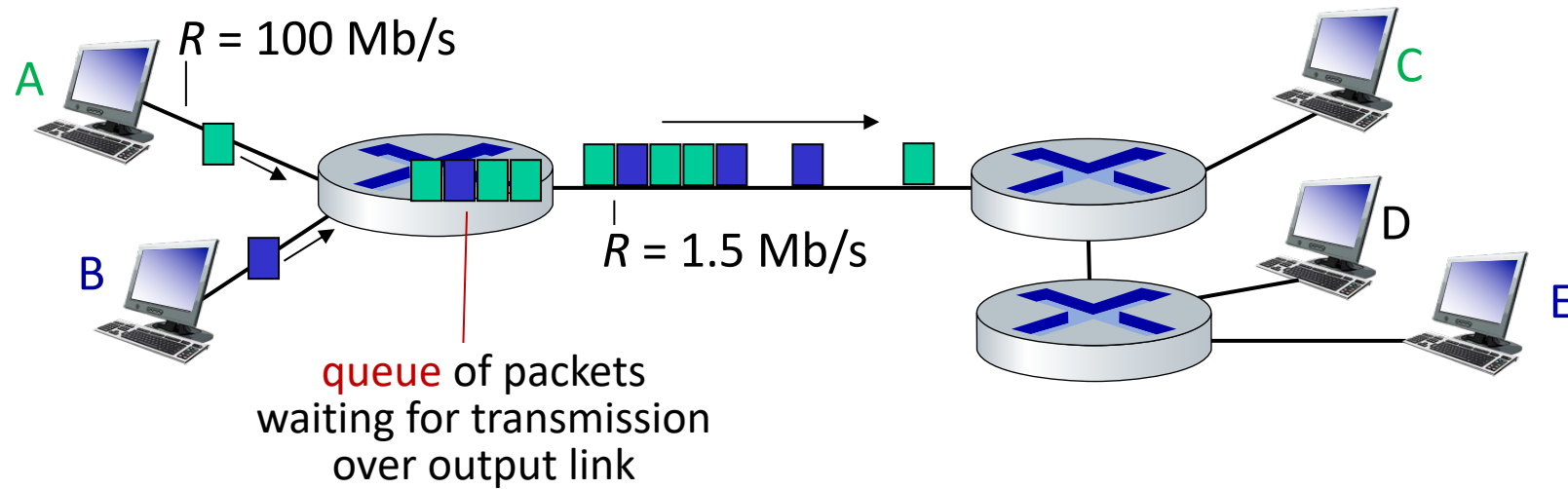
Bandwidth division into "pieces"
Dedicated allocation
Resource reservation



resource contention:

- ❑ aggregate resource demand can exceed amount available
- ❑ congestion: packets queue, wait for link use
- ❑ store and forward: packets move one hop at a time
 - ❖ Node receives complete packet before forwarding

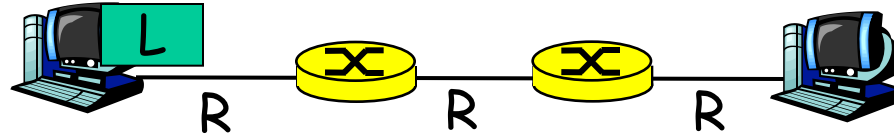
Packet-switching: queueing



Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Packet-switching: store-and-forward



- ❑ Takes L/R seconds to transmit (push out) packet of L bits on to link at R bps
- ❑ Entire packet must arrive at router before it can be transmitted on next link:
store and forward
- ❑ delay = $3L/R$ (assuming zero propagation delay)

} more on delay shortly ...

Example:

- ❑ $L = 7.5$ Mbits
- ❑ $R = 1.5$ Mbps
- ❑ delay = 15 sec

Packet switching versus circuit switching

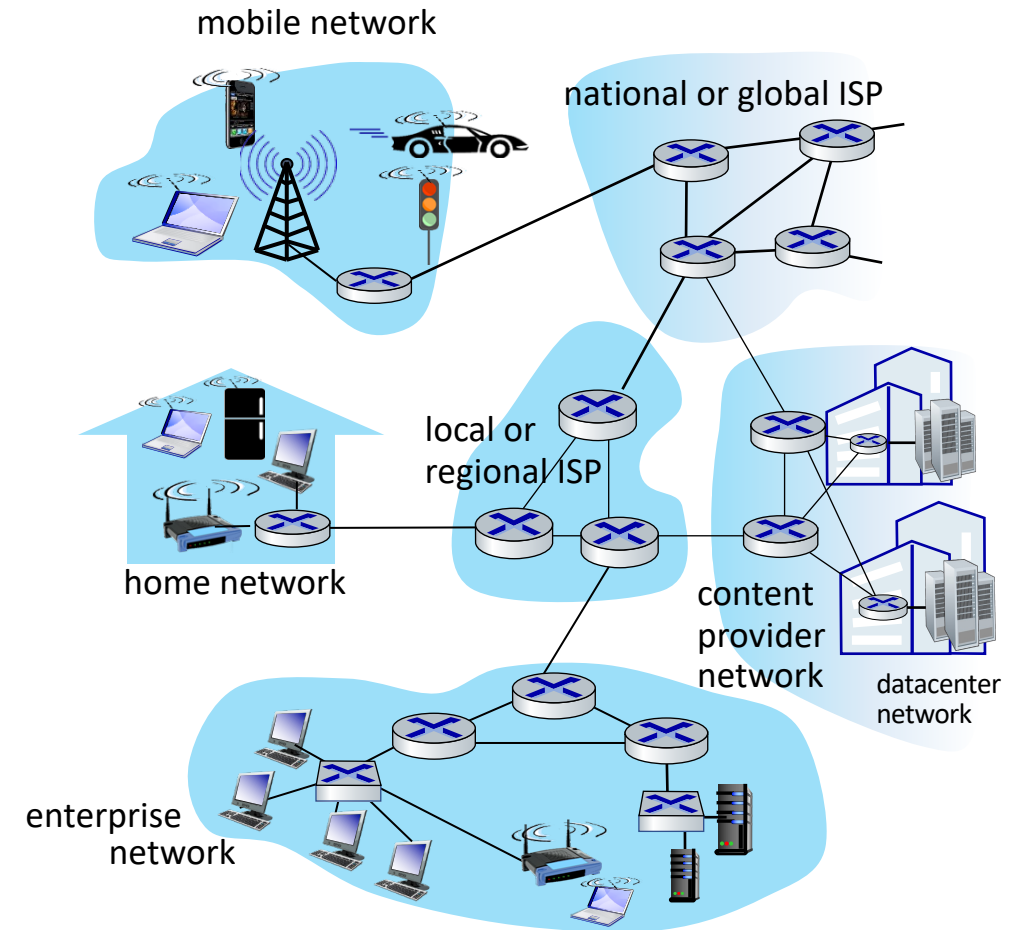
Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior with packet-switching?**
 - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

Internet structure: a "network of networks"

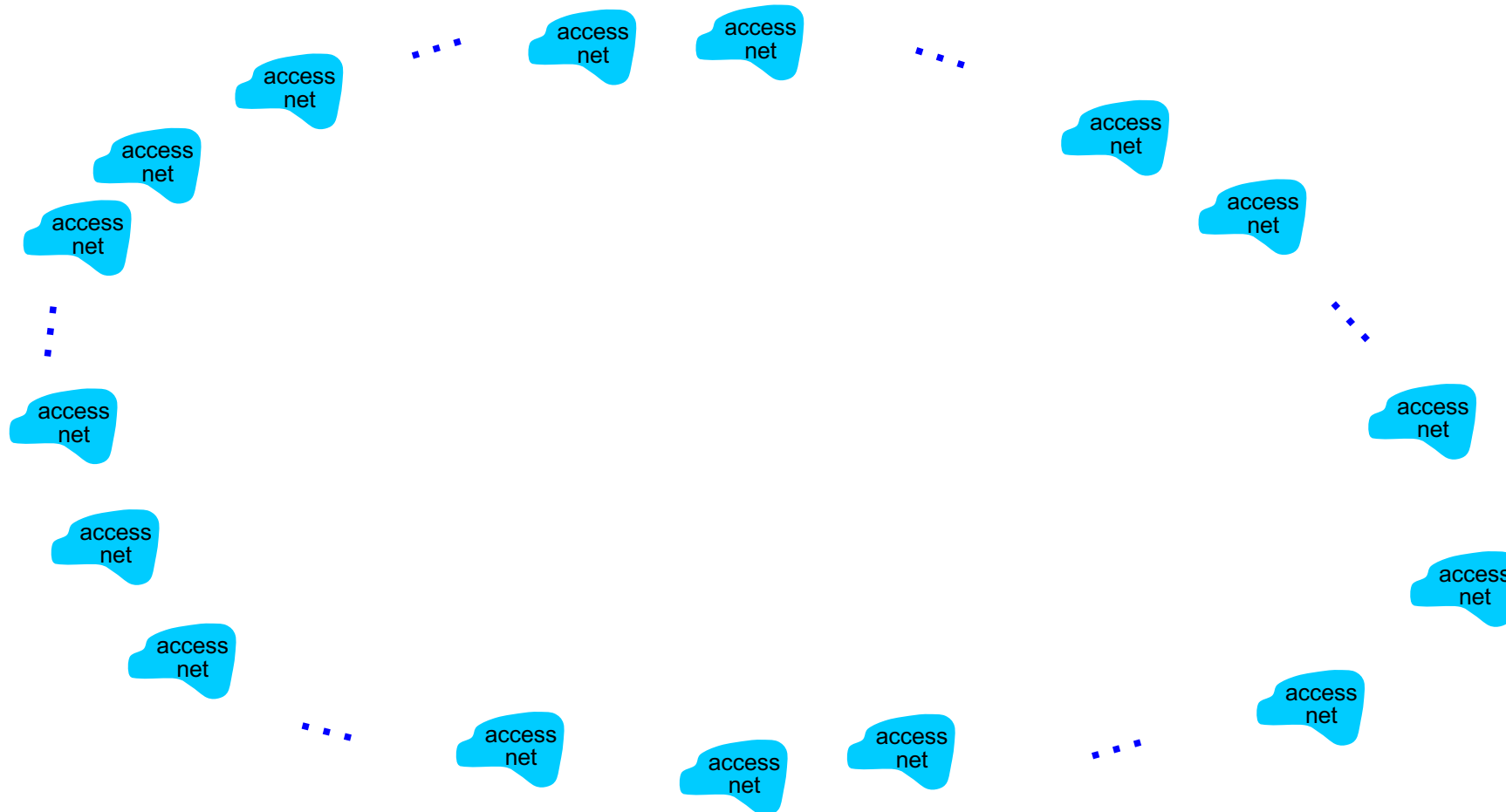
- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that *any* two hosts (*anywhere!*) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics**, **national policies**



Let's take a stepwise approach to describe current Internet structure

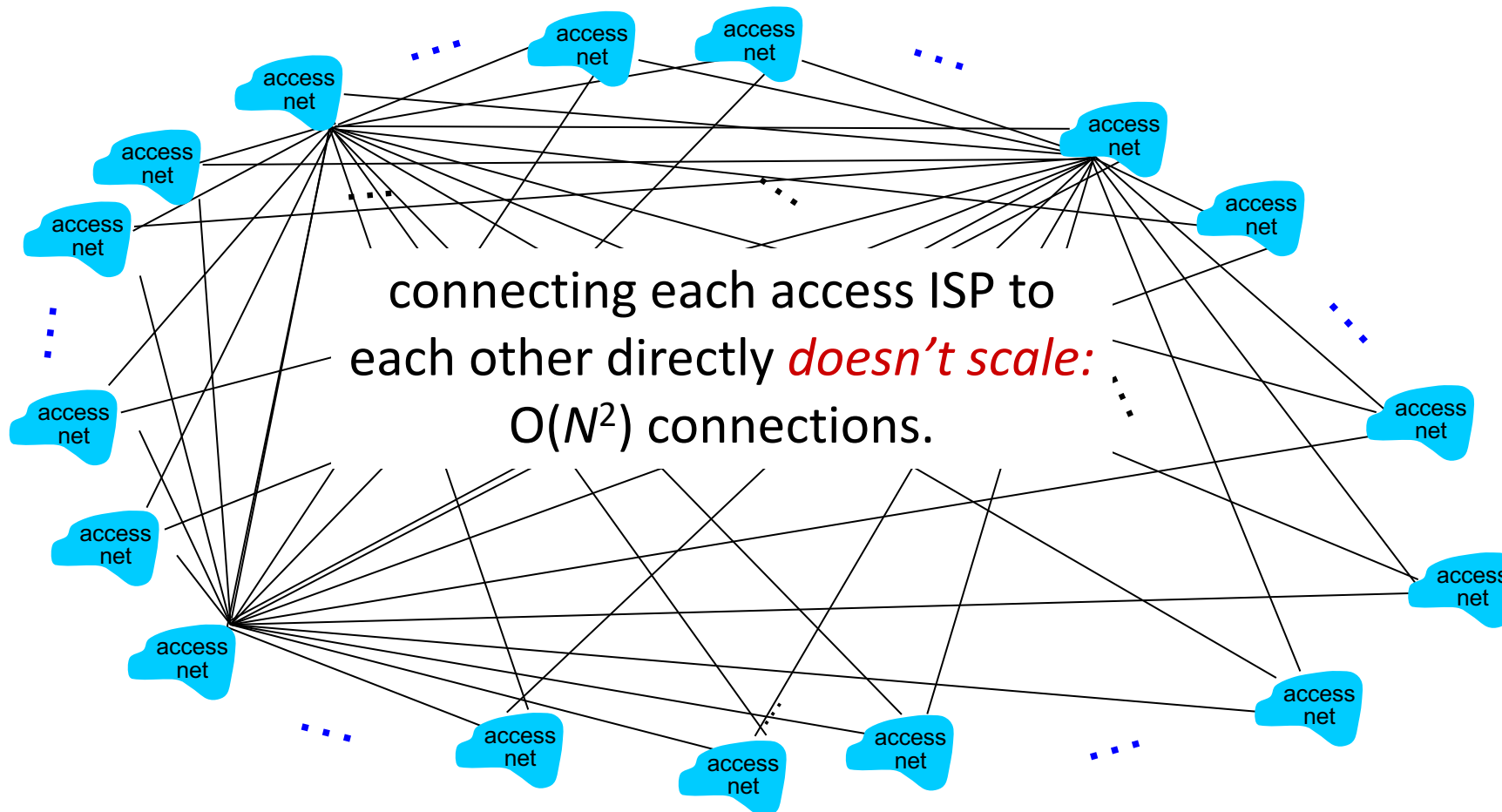
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

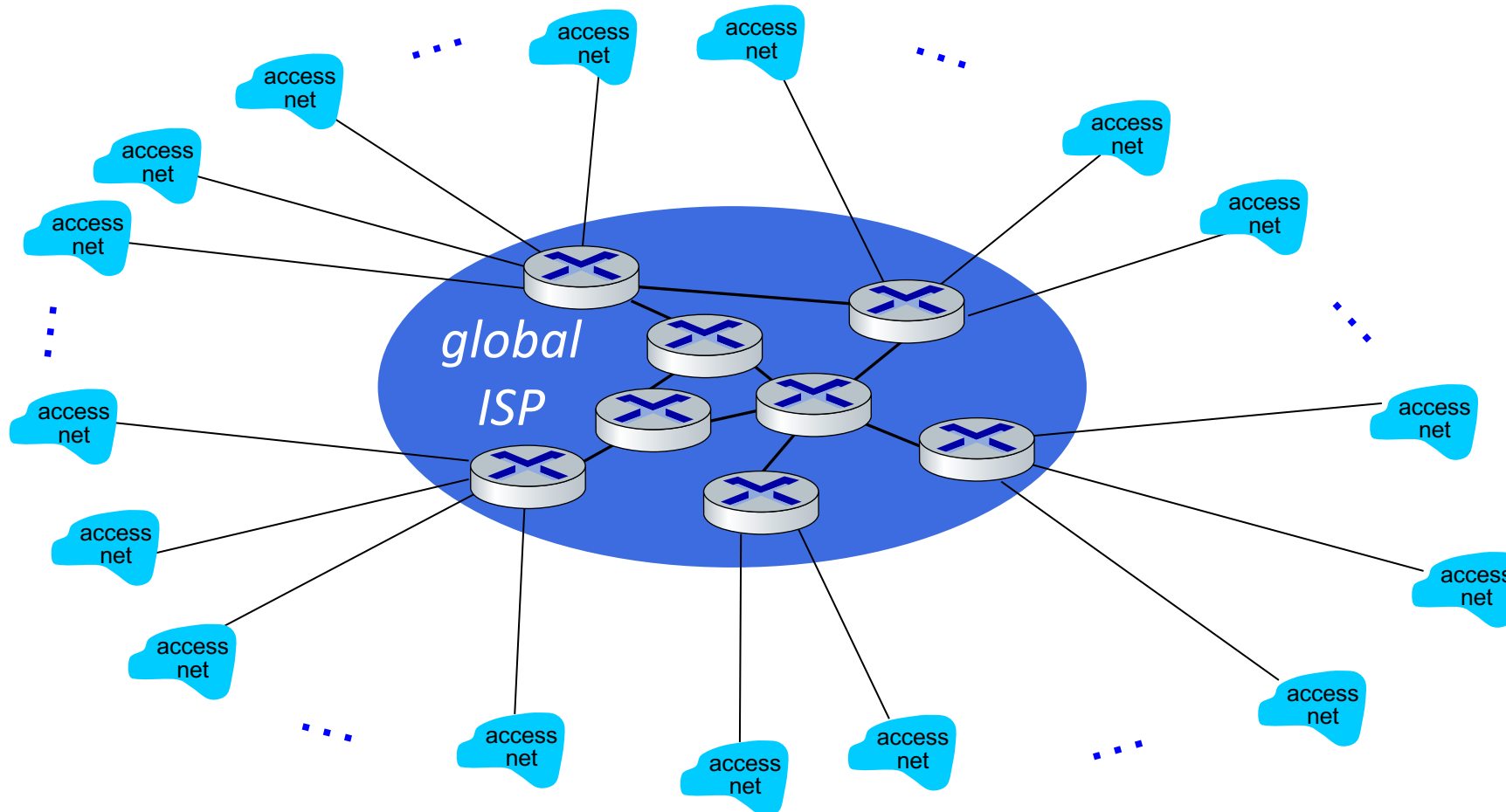
Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

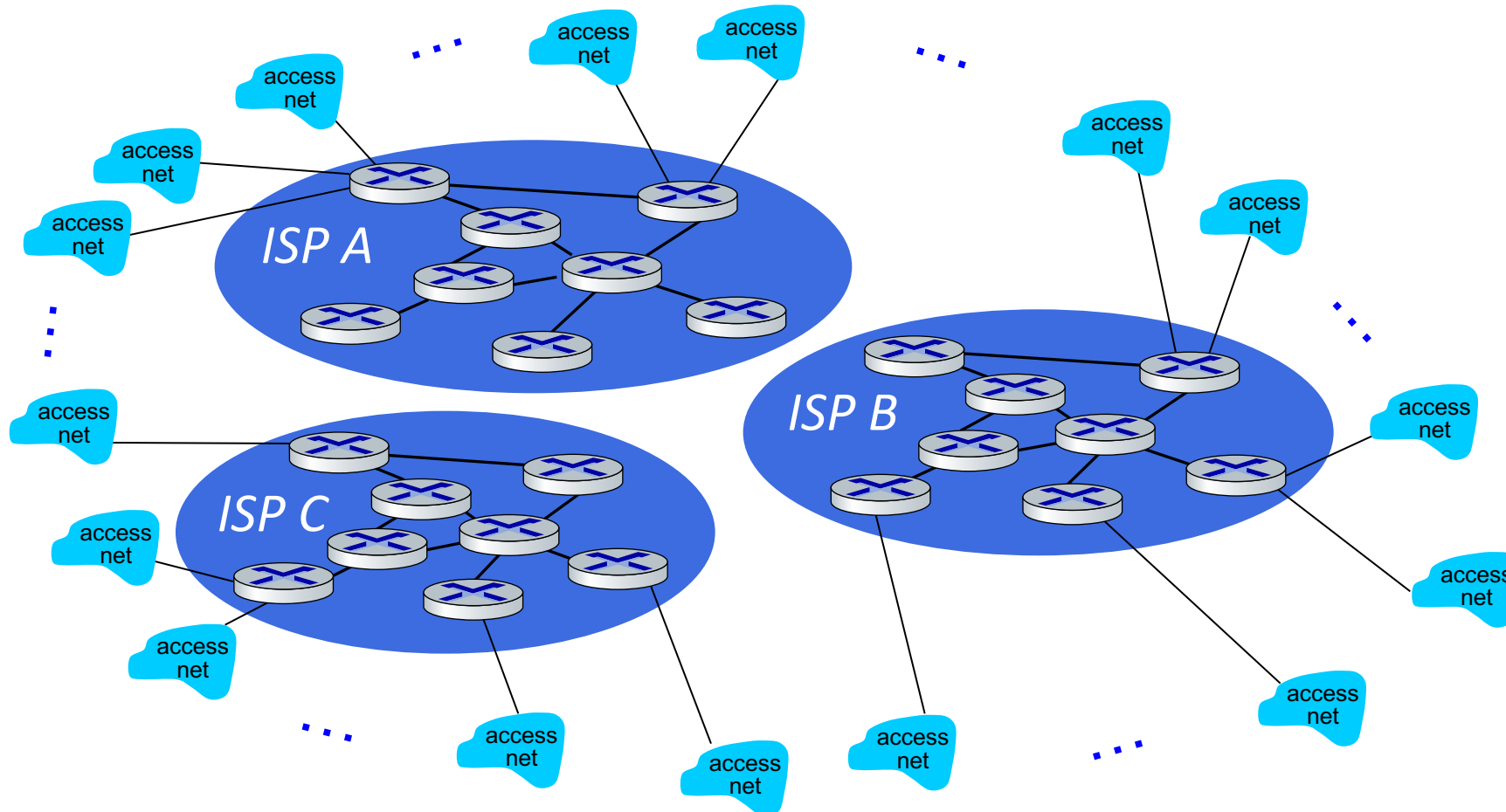
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



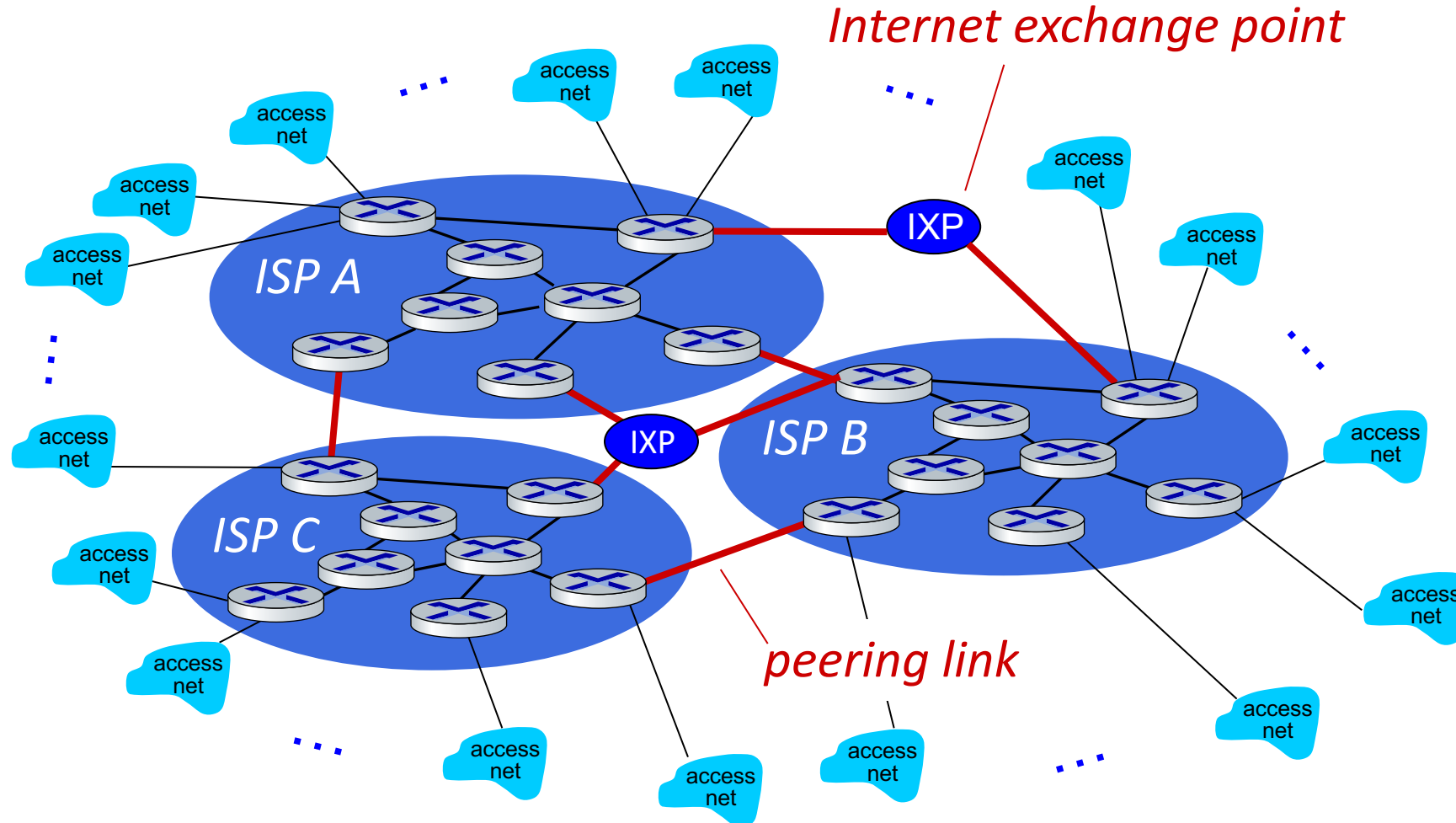
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors



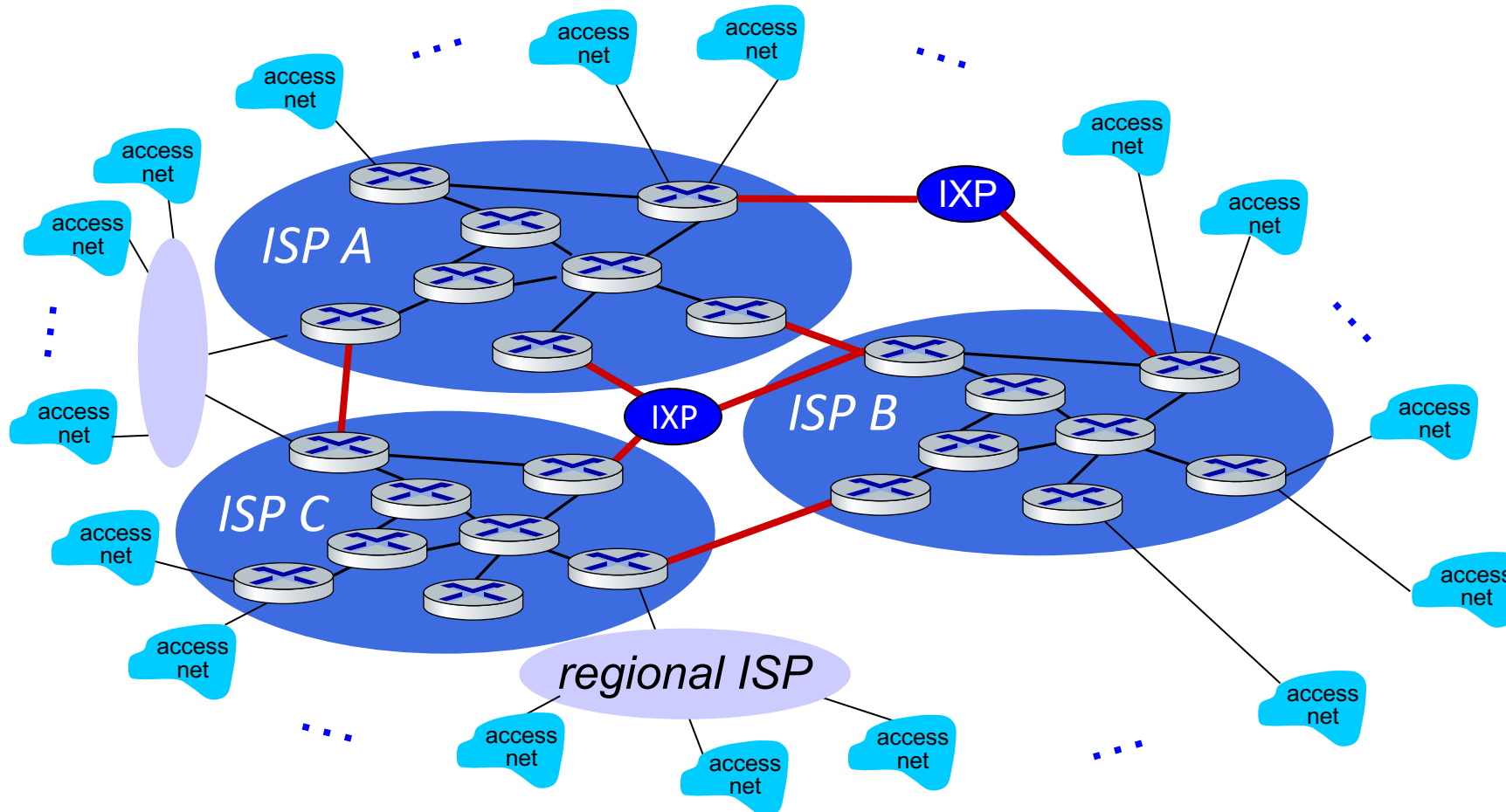
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors who will want to be connected



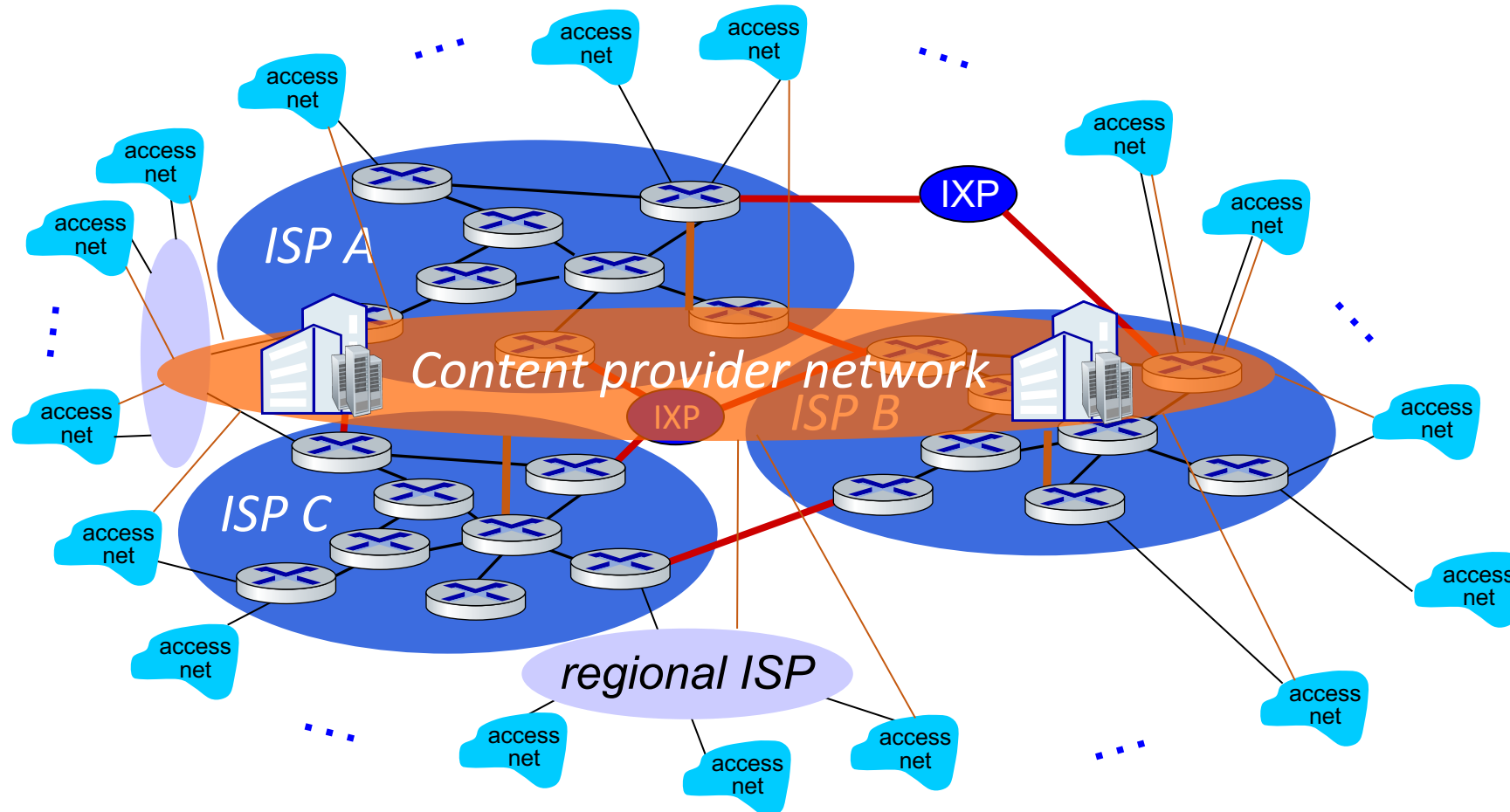
Internet structure: a "network of networks"

... and regional networks may arise to connect access nets to ISPs

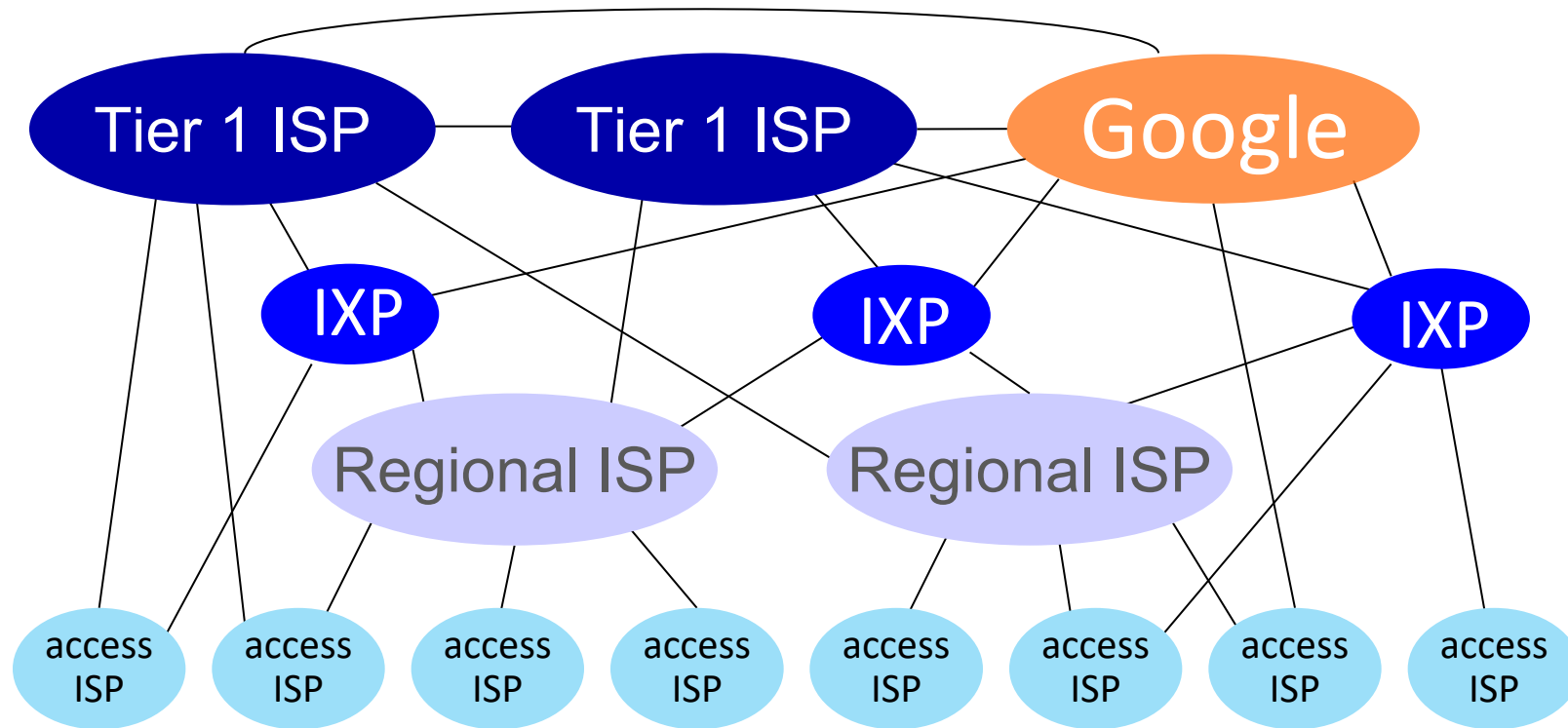


Internet structure: a "network of networks"

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

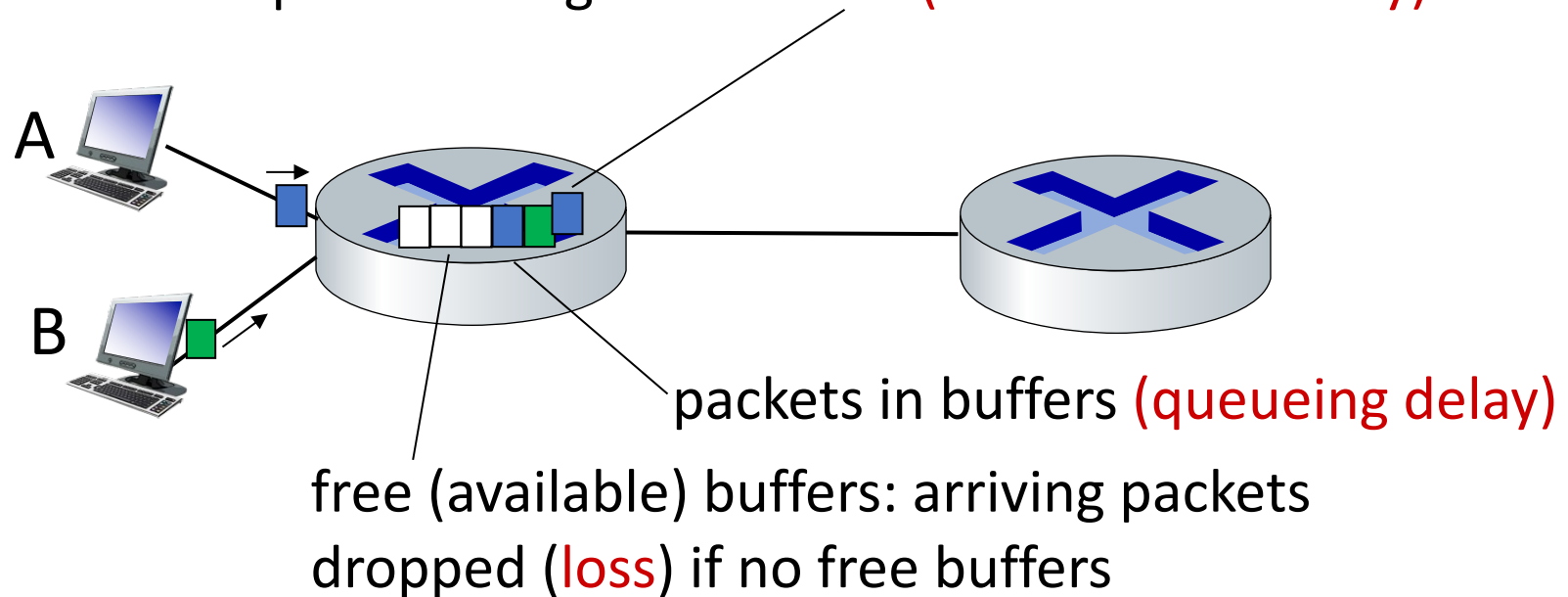
- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Chapter 1: roadmap

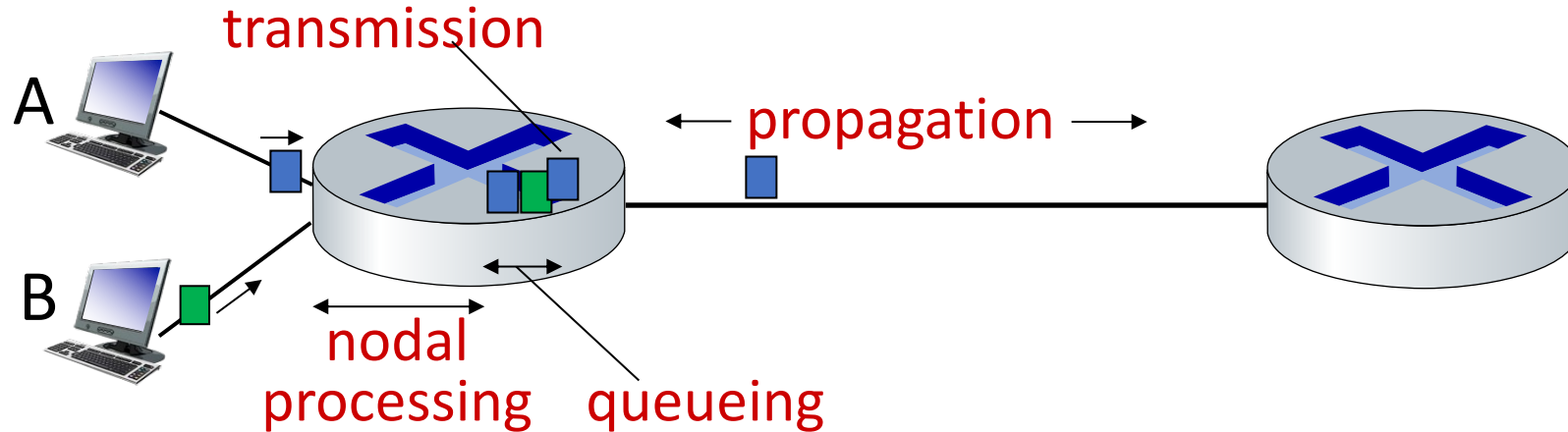
- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- **Performance: loss, delay, throughput**
- Security
- Protocol layers, service models
- History

How do packet delay and loss occur?

- packets *queue* in router buffers, waiting for turn for transmission
 - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet *loss* occurs when memory to hold queued packets fills up
 - packet being transmitted (*transmission delay*)



Packet delay: four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

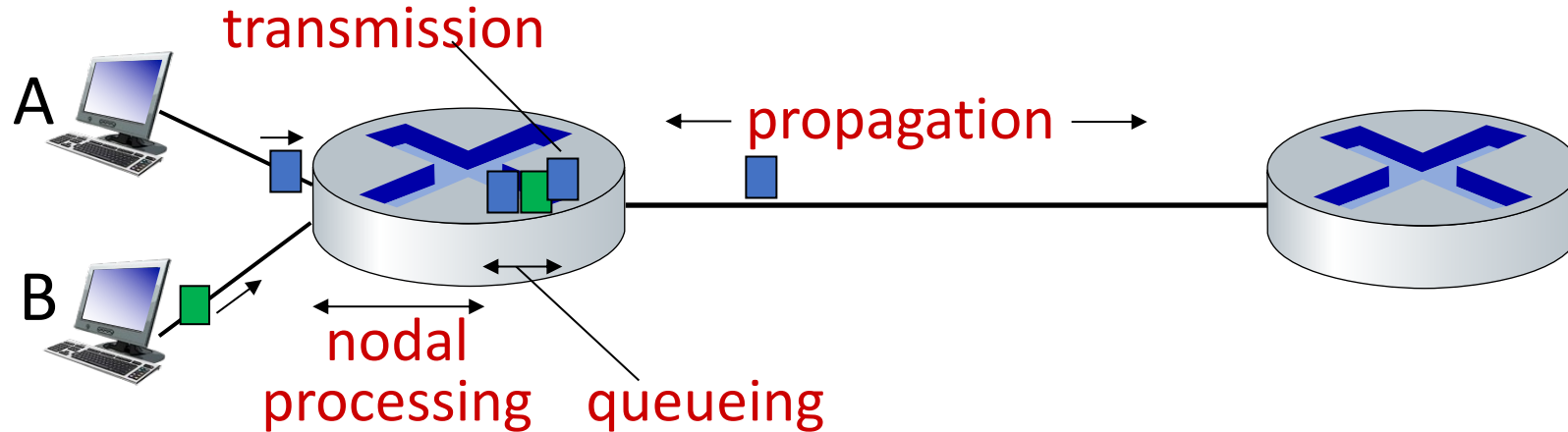
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet delay: four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link transmission rate (bps)

▪ $d_{\text{trans}} = L/R$

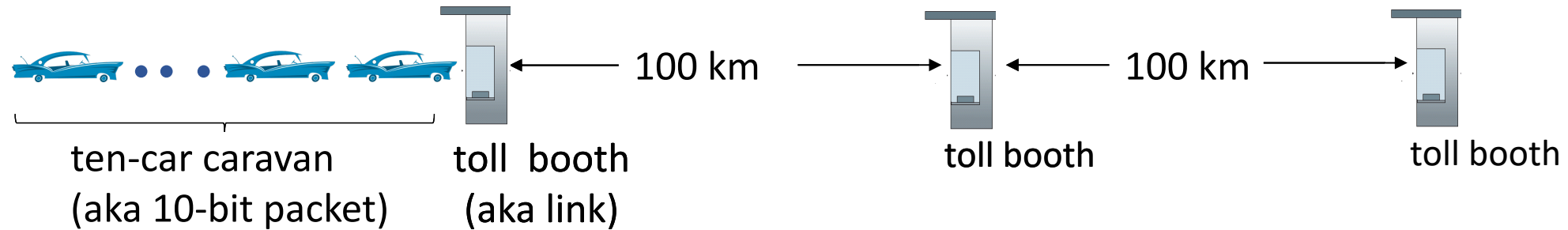
d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

▪ $d_{\text{prop}} = d/s$

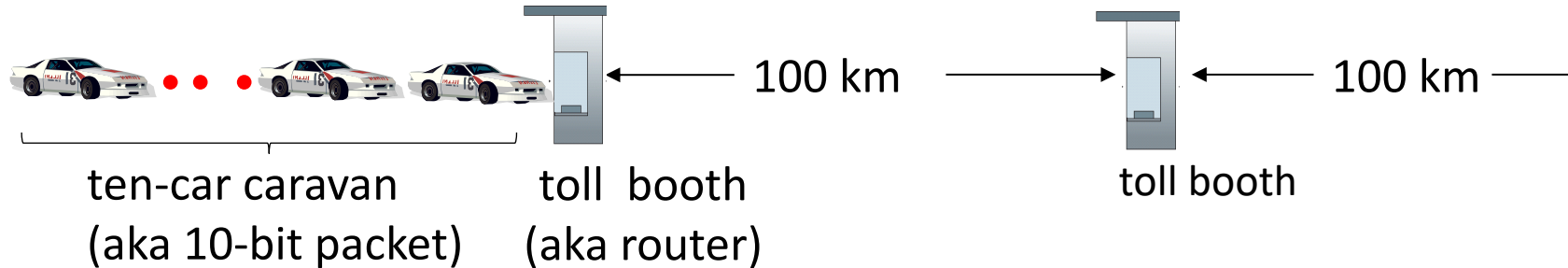
d_{trans} and d_{prop}
very different

Caravan analogy



- car $\sim$ bit; caravan $\sim$ packet; toll service $\sim$ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway = $12 * 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**

Caravan analogy



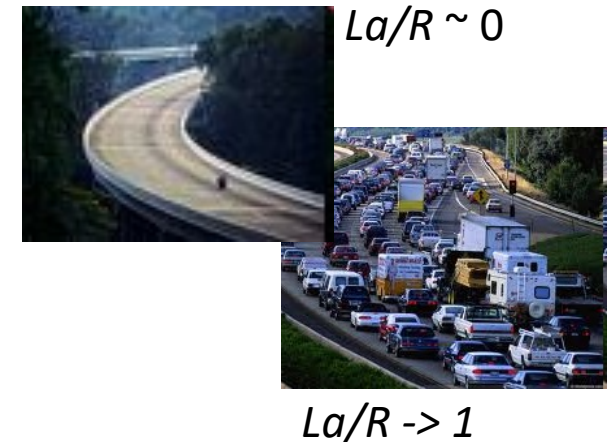
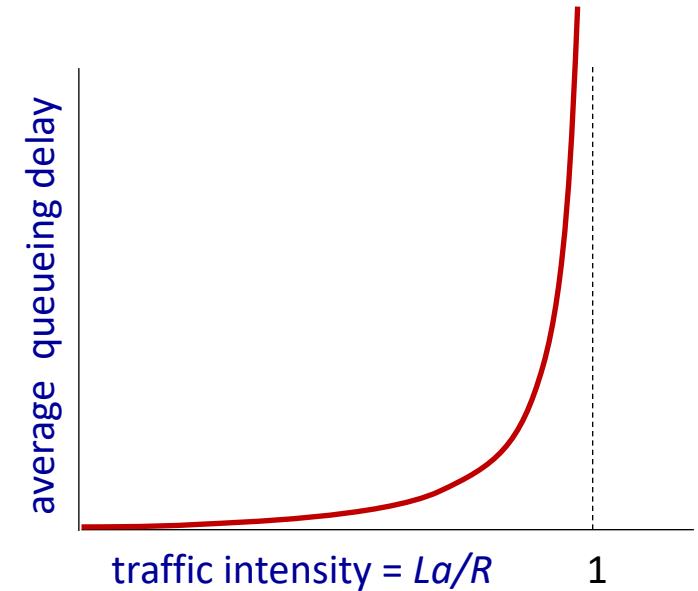
- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**
A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

Packet queueing delay (revisited)

- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

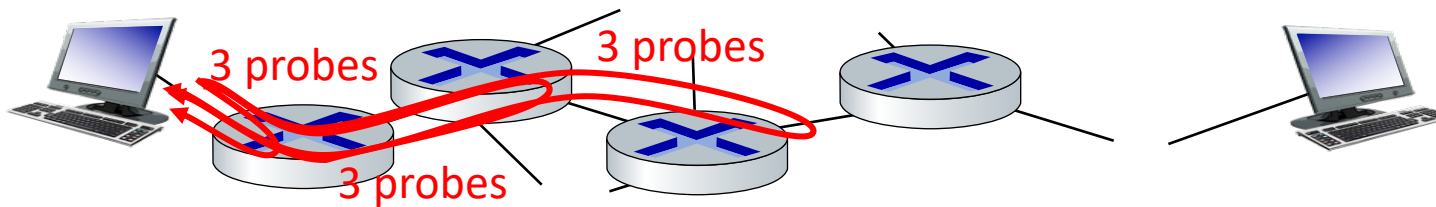
$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$

- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!



“Real” Internet delays, routes

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



“Real” Internet delays, routes

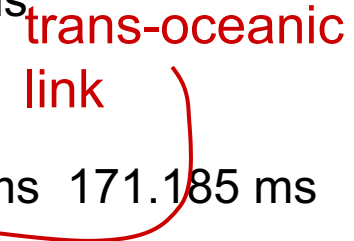
traceroute: .shanghai Unicom to www.usc.edu

3 delay measurements from
Fudan to ww.usc.edu

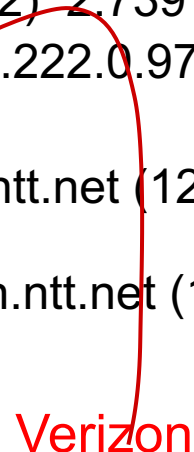


1 10.51.207.185 (10.51.207.185) 1.870 ms 1.344 ms 1.134 ms
2 10.51.1.2 (10.51.1.2) 1.152 ms 1.095 ms 1.057 ms
3 210.13.73.185 (210.13.73.185) 14.952 ms 2.684 ms 4.454 ms
4 * * *
5 112.64.252.190 (112.64.252.190) 3.431 ms 6.079 ms 2.142 ms
6 218.105.2.149 (218.105.2.149) 4.693 ms 5.245 ms 3.858 ms
7 218.105.10.22 (218.105.10.22) 2.739 ms 2.710 ms 2.678 ms
8 pos3-0.ig1.lax7.alter.net (208.222.0.97) 157.196 ms 166.491 ms 171.185 ms
9 * * *
10 ae-7.r01.lsanca20.us.bb.gin.ntt.net (129.250.8.85) 168.629 ms 190.266 ms
177.837 ms
11 ae-17.r01.lsanca07.us.bb.gin.ntt.net (129.250.4.207) 179.043 ms 179.474 ms
180.199 ms

trans-oceanic
link



Verizon



“Real” Internet delays, routes

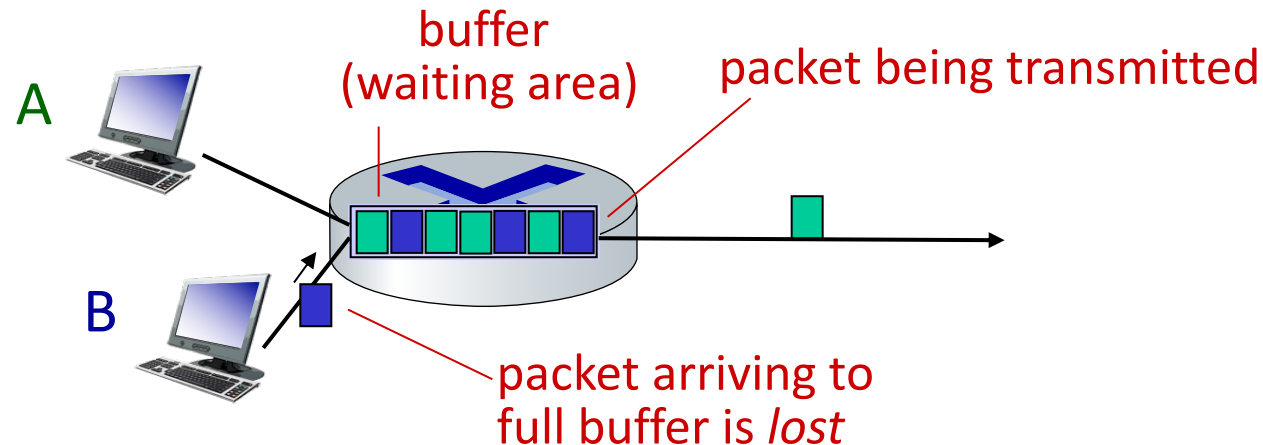
traceroute: shanghai unicom to www.usc.edu

```
11 ae-17.r01.lsanca07.us.bb.gin.ntt.net (129.250.4.207) 179.043 ms 179.474
    ms 180.199 ms
12 ntt-los-nettos-usc.ln.net (165.254.21.242) 175.243 ms 169.510 ms
    170.251 ms
13 130.152.182.83 (130.152.182.83) 159.538 ms 169.551 ms 192.469 ms
14 fw6-rtr-border-cal.usc.edu (128.125.251.227) 194.779 ms 162.290 ms
    158.598 ms
15 rtr6-fw6.usc.edu (128.125.255.146) 160.823 ms 161.753 ms 159.159 ms
16 v249-gw-33.usc.edu (68.181.194.69) 160.257 ms 158.741 ms 164.976
    ms
17 fwc01.usc.edu (128.125.255.190) 161.820 ms 161.706 ms 164.768 ms
18 * * *
19 cwis.usc.edu (128.125.253.146) 166.461 ms 158.701 ms 163.325 ms
```

* means no response (probe lost, router not replying)

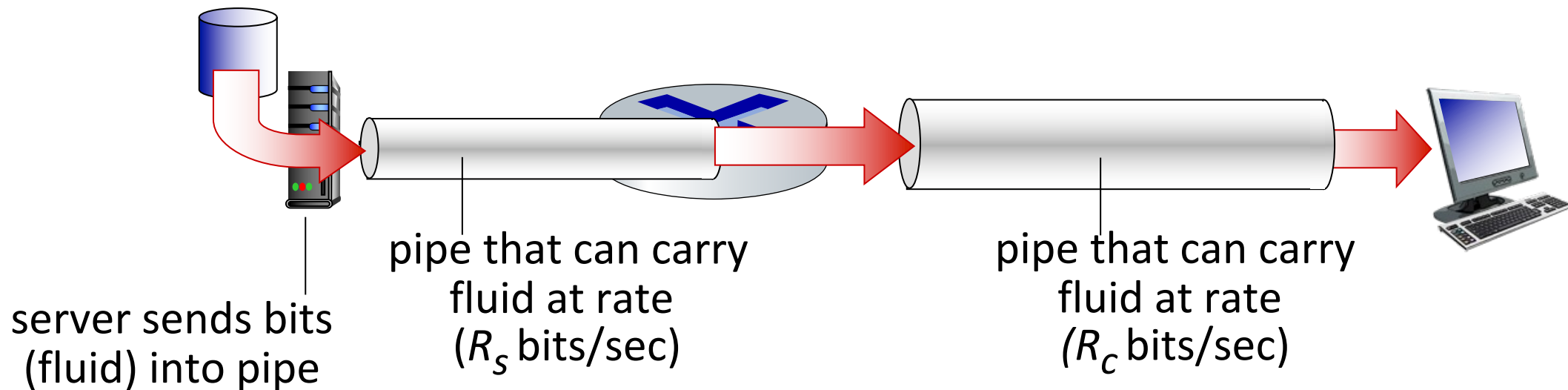
Packet loss

- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



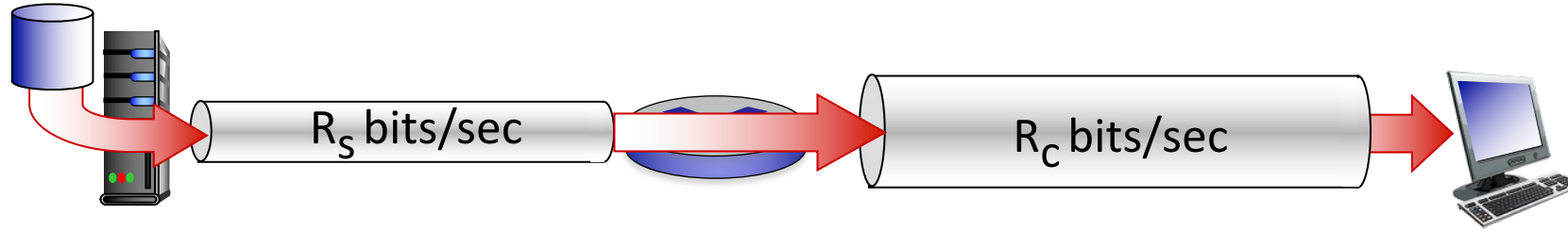
Throughput

- *throughput*: rate (bits/time unit) at which bits are being sent from sender to receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over longer period of time

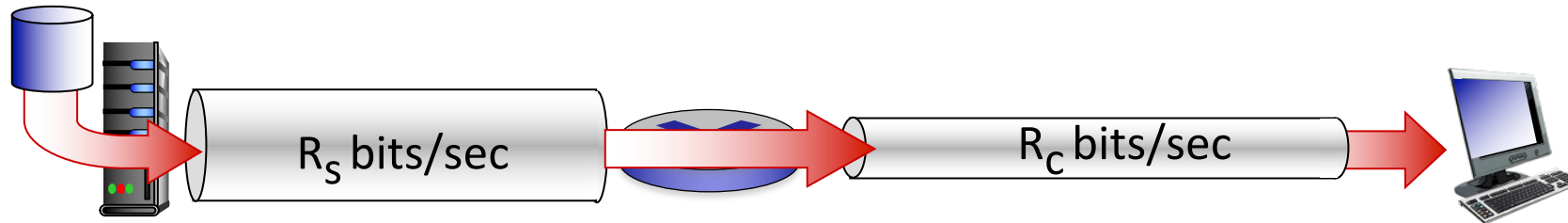


Throughput

$R_s < R_c$ What is average end-end throughput?



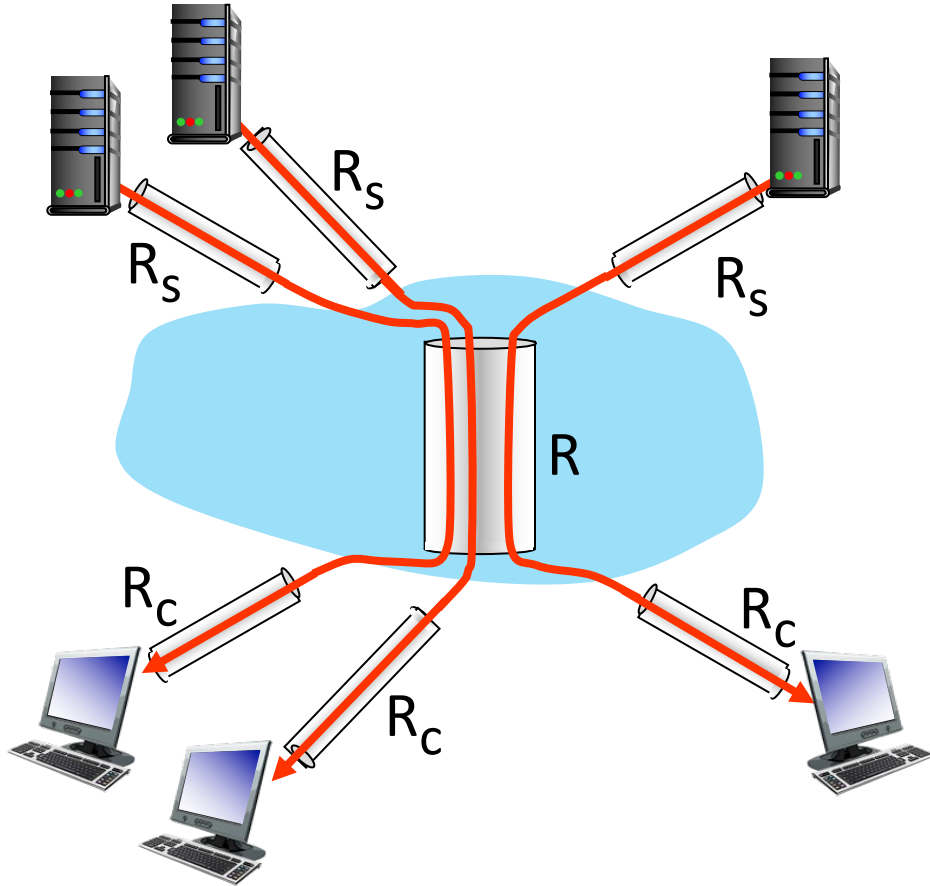
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Throughput: network scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- **Security**
- Protocol layers, service models
- History

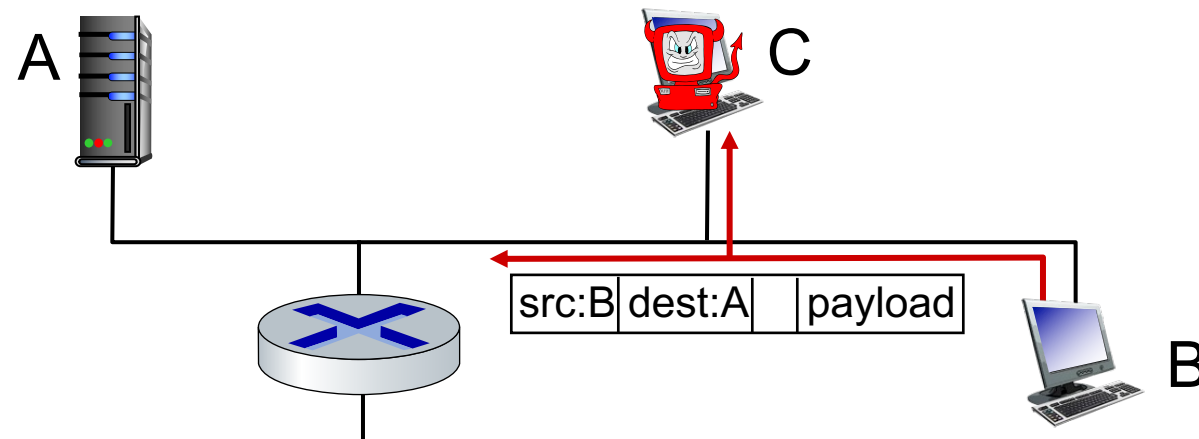
Network security

- Internet not originally designed with (much) security in mind
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!
- We now need to think about:
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks

Bad guys: packet interception

packet “sniffing”:

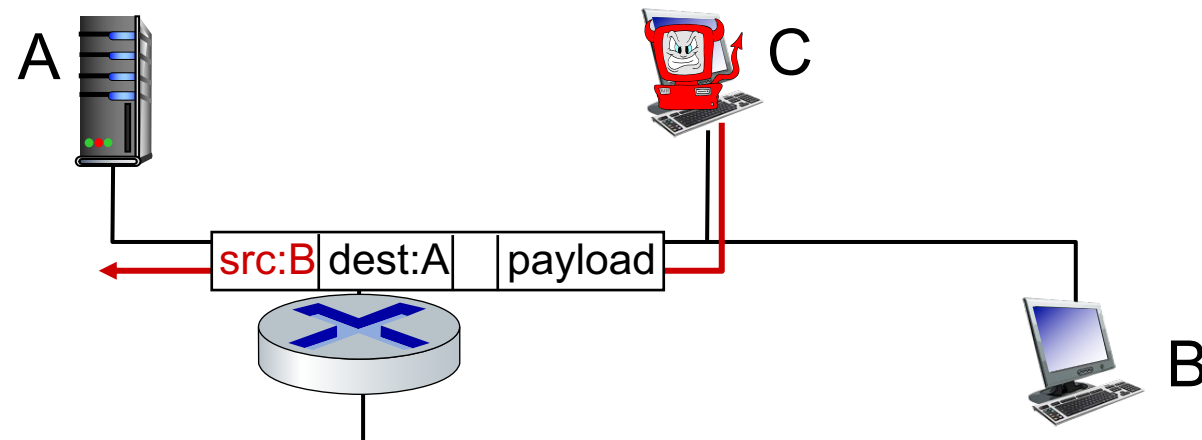
- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



Wireshark software used for our end-of-chapter labs is a (free) packet-sniffer

Bad guys: fake identity

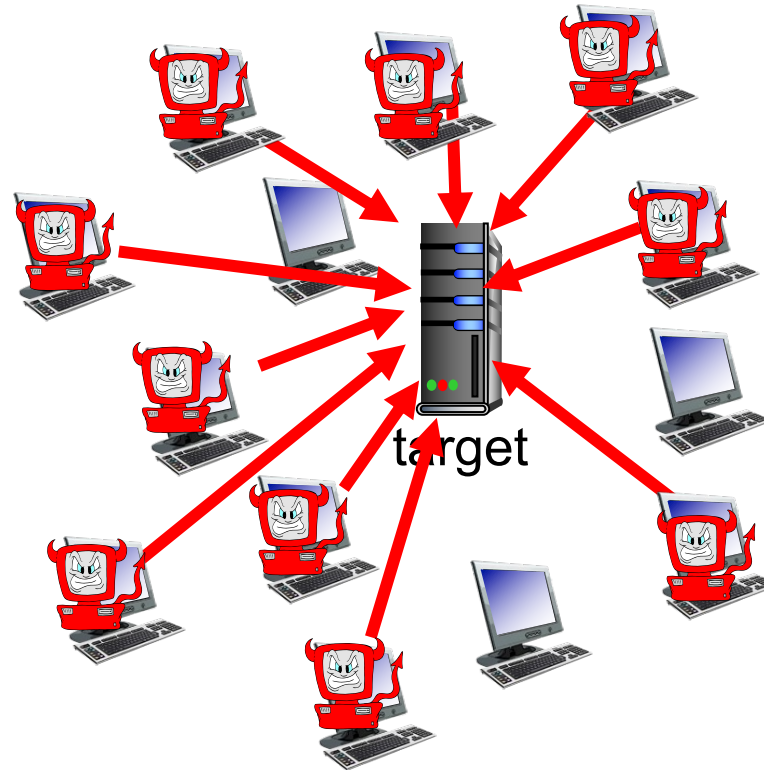
IP spoofing: injection of packet with false source address



Bad guys: denial of service

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Lines of defense:

- **authentication:** proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **confidentiality:** via encryption
- **integrity checks:** digital signatures prevent/detect tampering
- **access restrictions:** password-protected VPNs
- **firewalls:** specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks

Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- **Protocol layers, service models**
- History

Protocol “layers” and reference models

Networks are complex,
with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any
hope of *organizing*
structure of network?

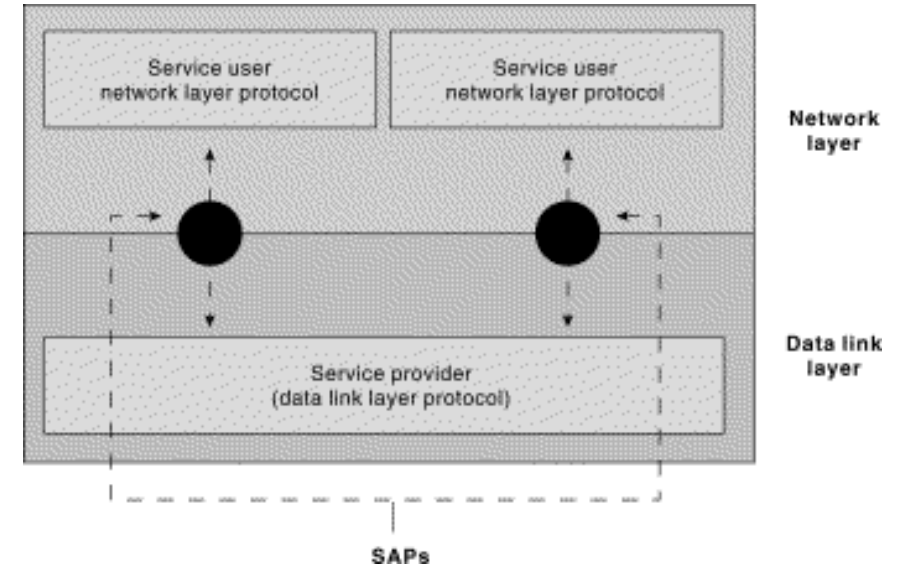
- and/or our *discussion*
of networks?

Why layering?

Approach to designing/discussing complex systems:

- explicit structure allows identification, relationship of system's pieces
 - layered *reference model* for discussion
- modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

Interfaces



❖ Each protocol offers an interface to its users, and expects one from the layers on which it builds

- Syntax and semantics strike again
 - Data formats
 - Interface characteristics, e.g. IP service model

❖ Protocols build upon each other

- Add value
 - E.g., a reliable protocol running on top of IP
- Reuse
 - E.g., OS provides TCP, so apps don't have to rewrite

Implementing Protocol

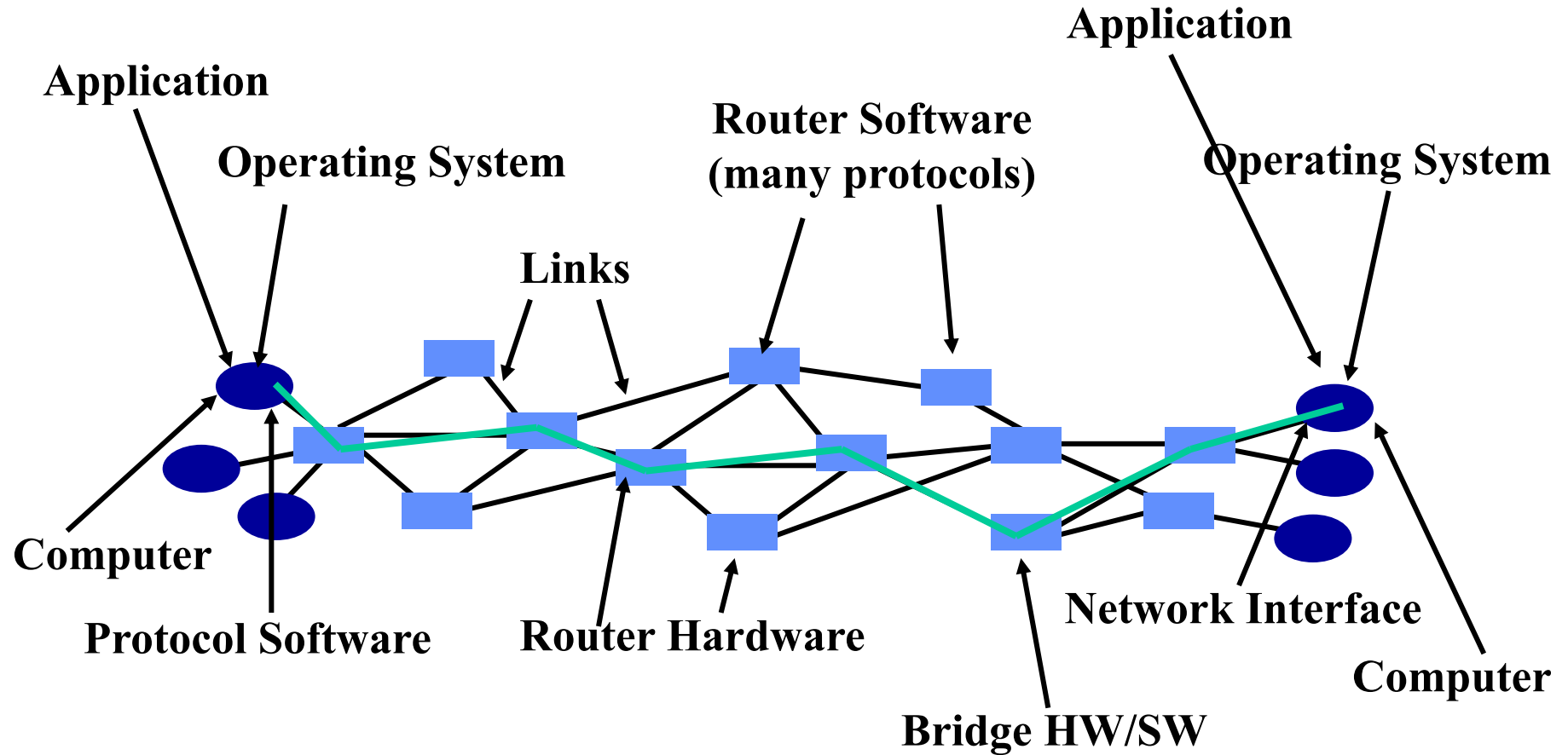
❖ Requirements for packets:

- Packet size limit?
- Header information: Addresses, etc.
- From where/who, to where ? Naming and address
 - POST, Telephone

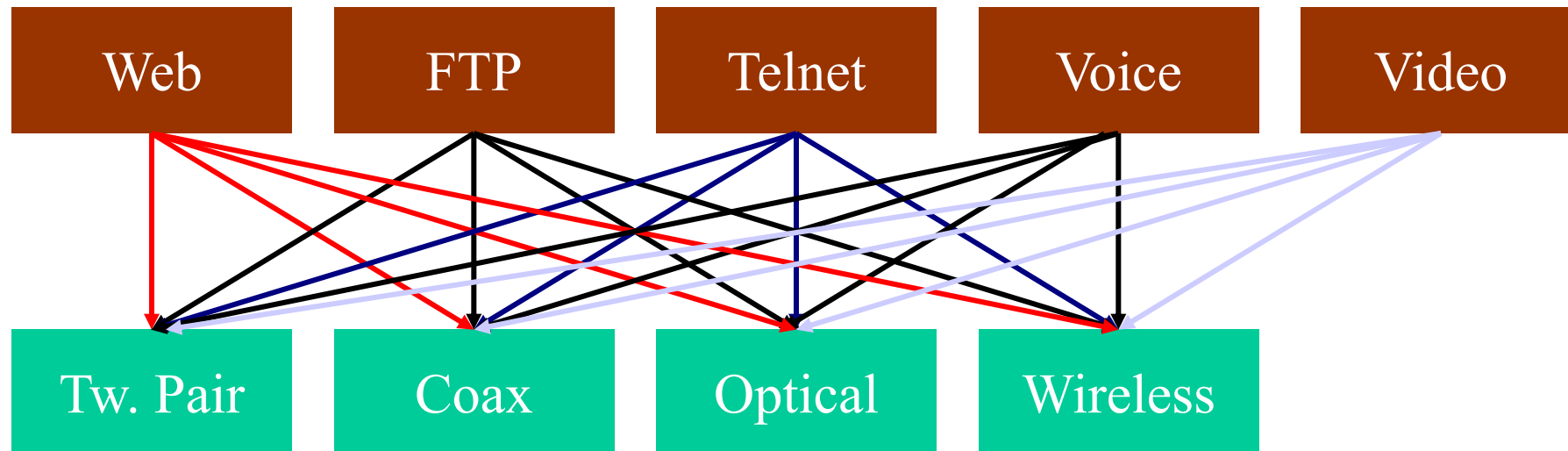
❖ Almost in all lectures

Q: The relationship between Data link layer name and IP name?

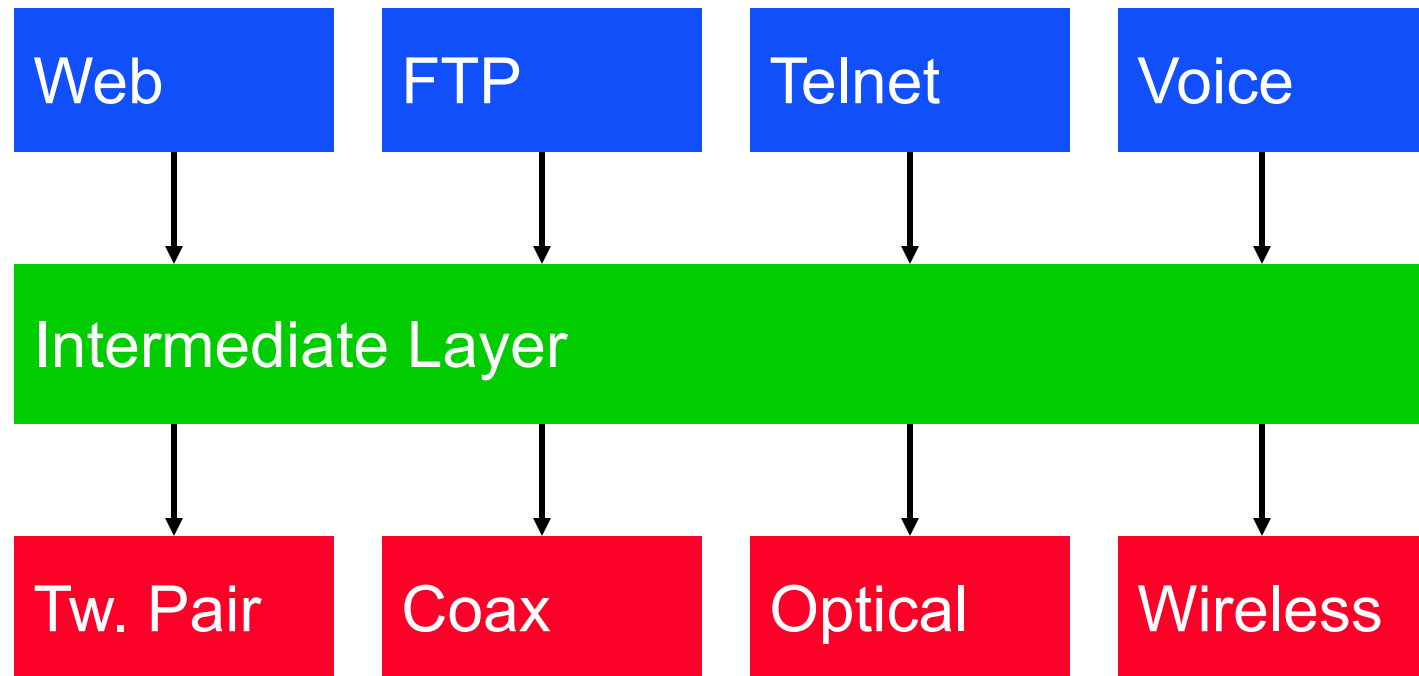
Too Many Network Components



Example: no abstraction



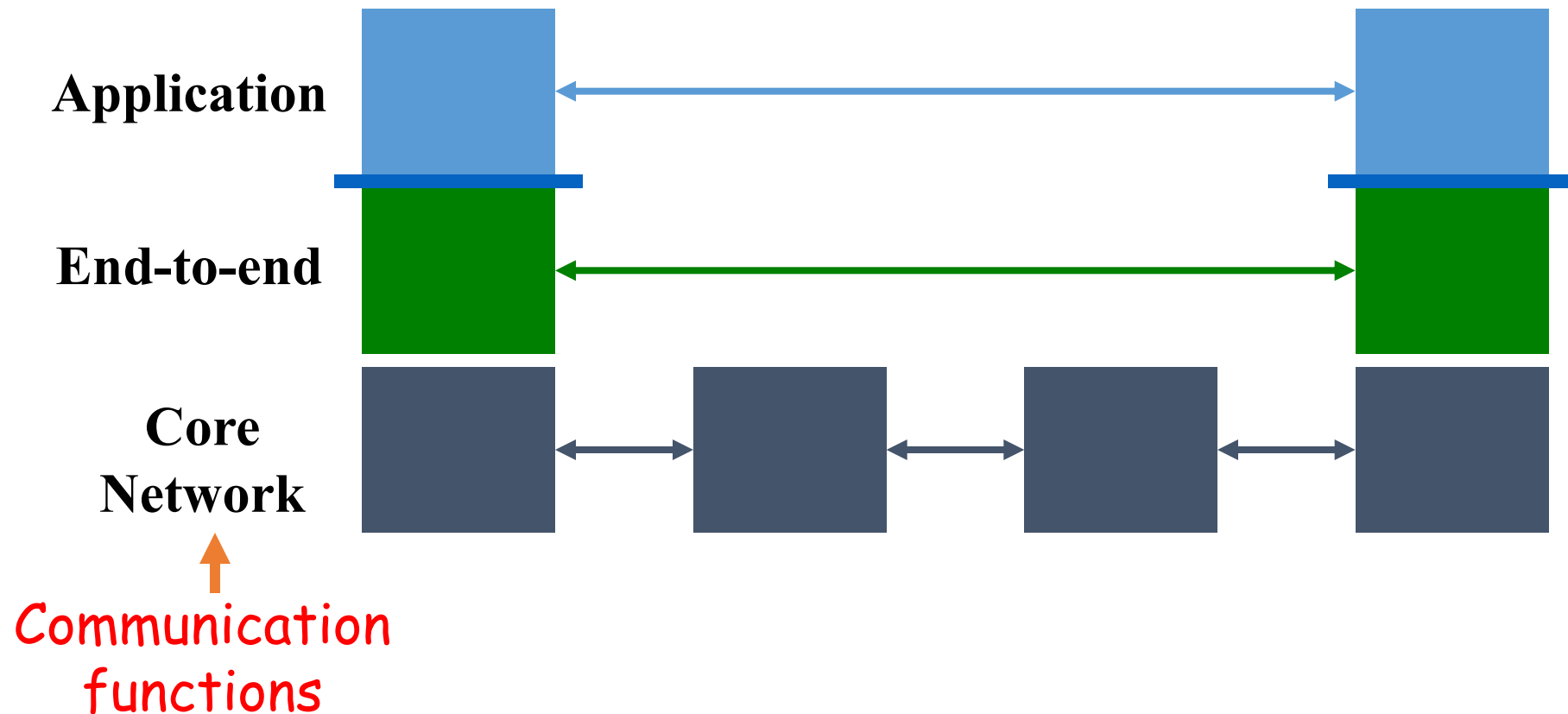
Three layers model : Hardware abstraction



Resources for Process | Communication Functions

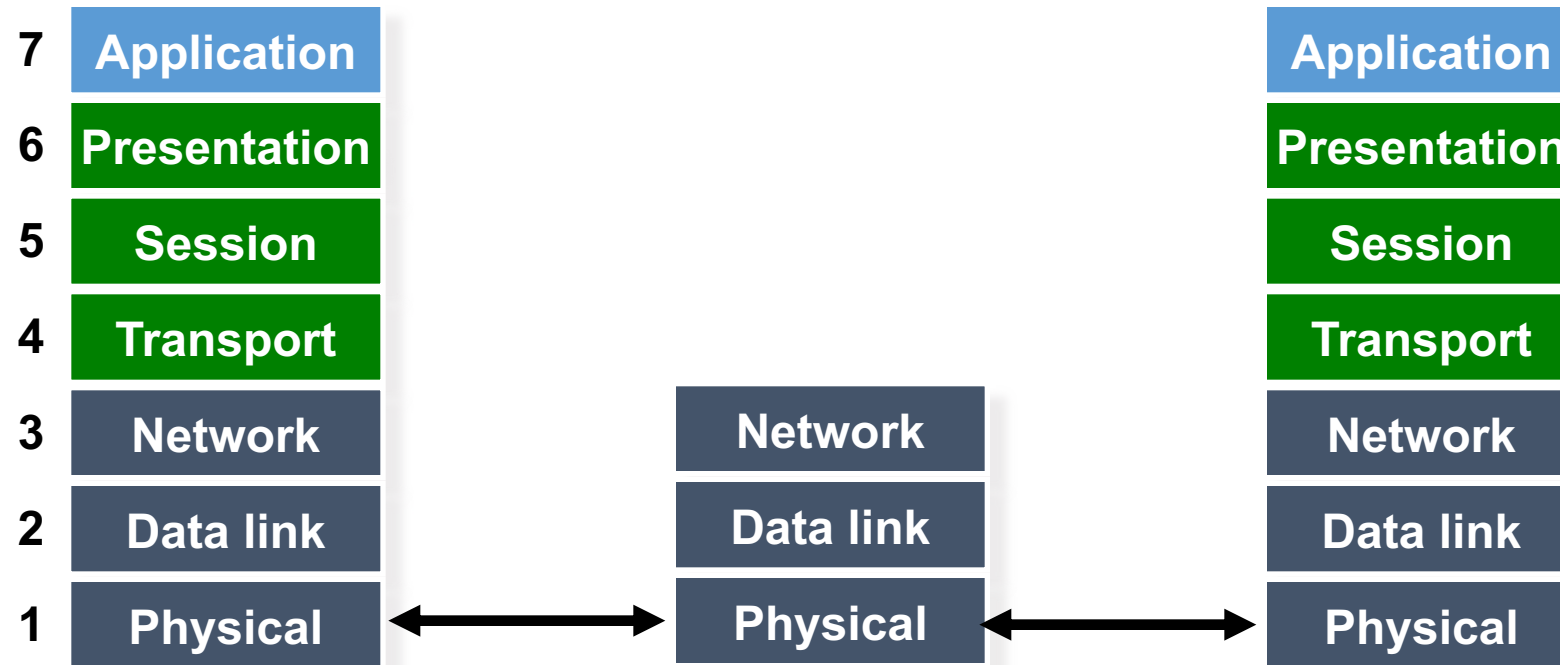
How to do abstractions?

End-to-end abstraction



A Layered Network Model

The Open Systems Interconnection (OSI) Model.



OSI Motivation

- ❖ Standard way of breaking up a system in a set of components, but the components are organized as a set of layers.
 - Only horizontal and vertical communication
 - Components/layers can be implemented and modified in isolation
- ❖ Each layer offers a service to the higher layer, using the services of the lower layer.
- ❖ "Peer" layers on different systems communicate via a protocol.
 - higher level protocols (e.g. TCP/IP, Appletalk) can run on multiple lower layers
 - multiple higher level protocols can share a single physical network
- ❖ "It's only a model!" - TCP/IP has been crazy successful, and it's not based on a rigid OSI model. But the OSI model has been very successful at shaping thought.

Advantages of layering

dealing with complex systems:

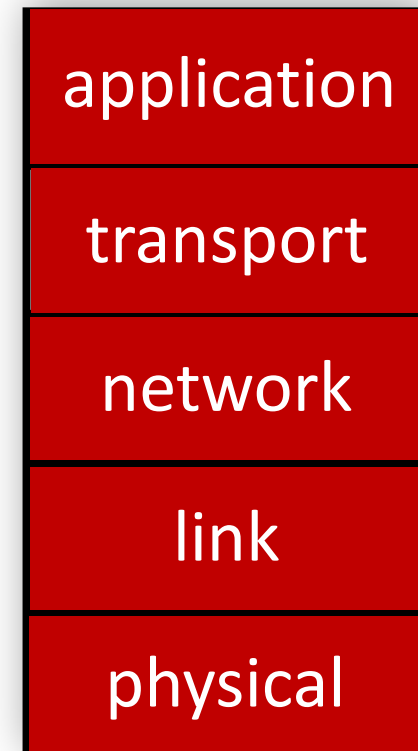
- ❖ explicit structure allows identification, relationship of complex system's pieces
 - layered *reference model* for discussion
- ❖ modularization eases maintenance, updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
- ❖ layering considered harmful?

Limitations of the Layered Model

- ❖ Some layers are not always cleanly separated.
 - Inter-layer dependencies in implementations for performance reasons
 - Some dependencies in the standards (header checksums)
- ❖ Higher layers not always well defined.
 - Session, presentation, application layers
- ❖ Lower layers have "sublayers".
 - Usually very well defined (e.g., LAN, SONET protocol)
- ❖ Interfaces are not really standardized.
 - It would be hard to mix and match layers from independent implementations, e.g., windows network apps on unix (without compatability library)
 - Many cross-layer assumptions, e.g. buffer management

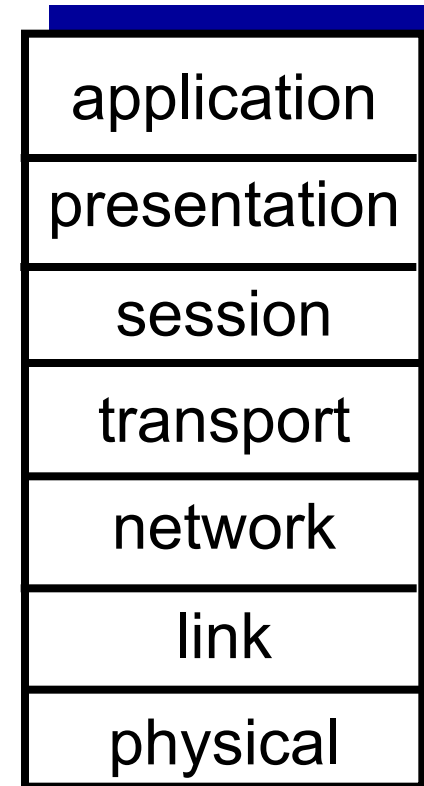
Layered Internet protocol stack

- *application*: supporting network applications
 - HTTP, IMAP, SMTP, DNS
- *transport*: process-process data transfer
 - TCP, UDP
- *network*: routing of datagrams from source to destination
 - IP, routing protocols
- *link*: data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- *physical*: bits "on the wire"



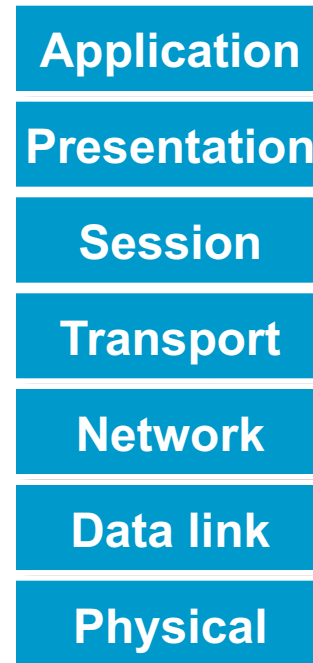
ISO/OSI reference model

- ❖ *presentation*: allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- ❖ *session*: synchronization, checkpointing, recovery of data exchange
- ❖ Internet stack “missing” these layers!
 - these services, *if needed*, must be implemented in application
 - needed?



Different Sources of Components

- ❖ Application: web server/browser, mail, distributed game,...
- ❖ Presentation/session.
 - Often part of application
 - Sometimes a library
- ❖ Transport/network.
 - Typically part of the operating system
- ❖ Datalink.
 - Often written by vendor of the network interface hardware
- ❖ Physical.
 - Hardware: card and link



The TCP/IP Model

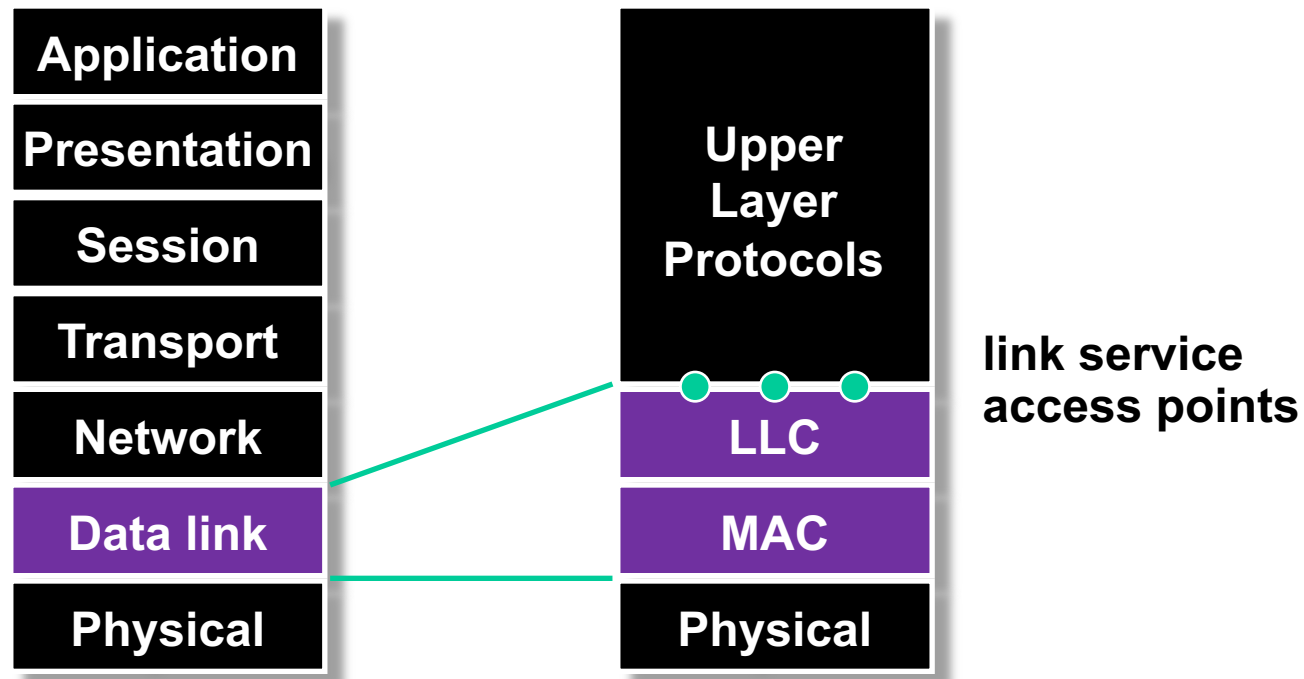
De facto industrial standard:

TCP/IP + IEEE802 link layer

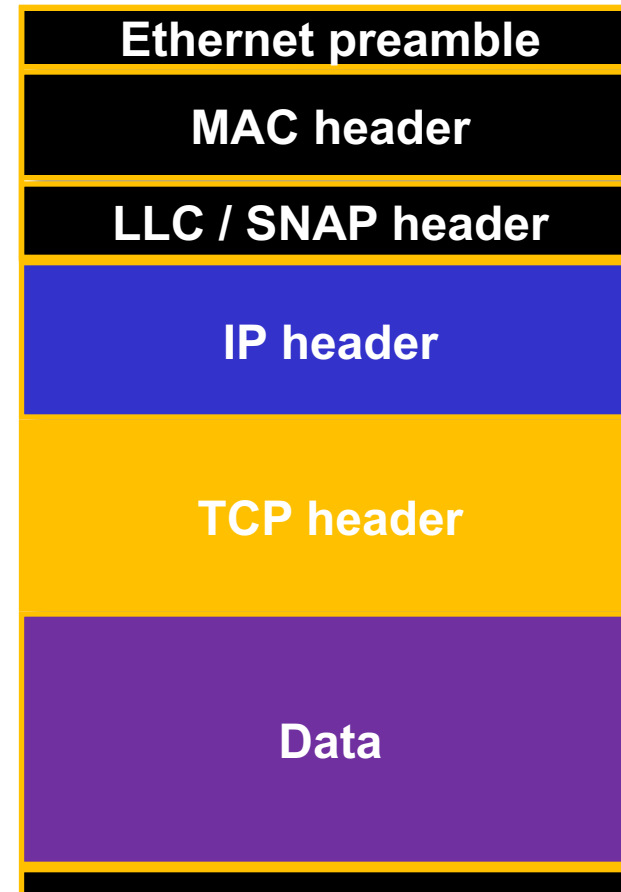
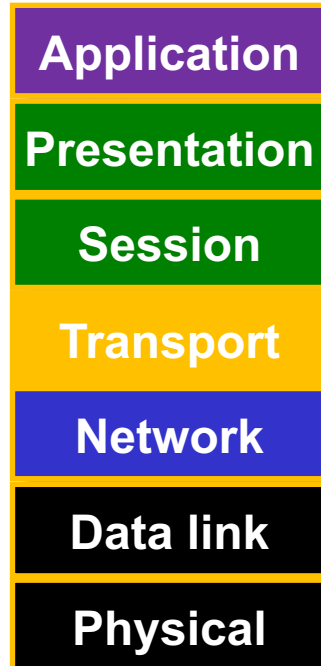


Local Area Network Protocols

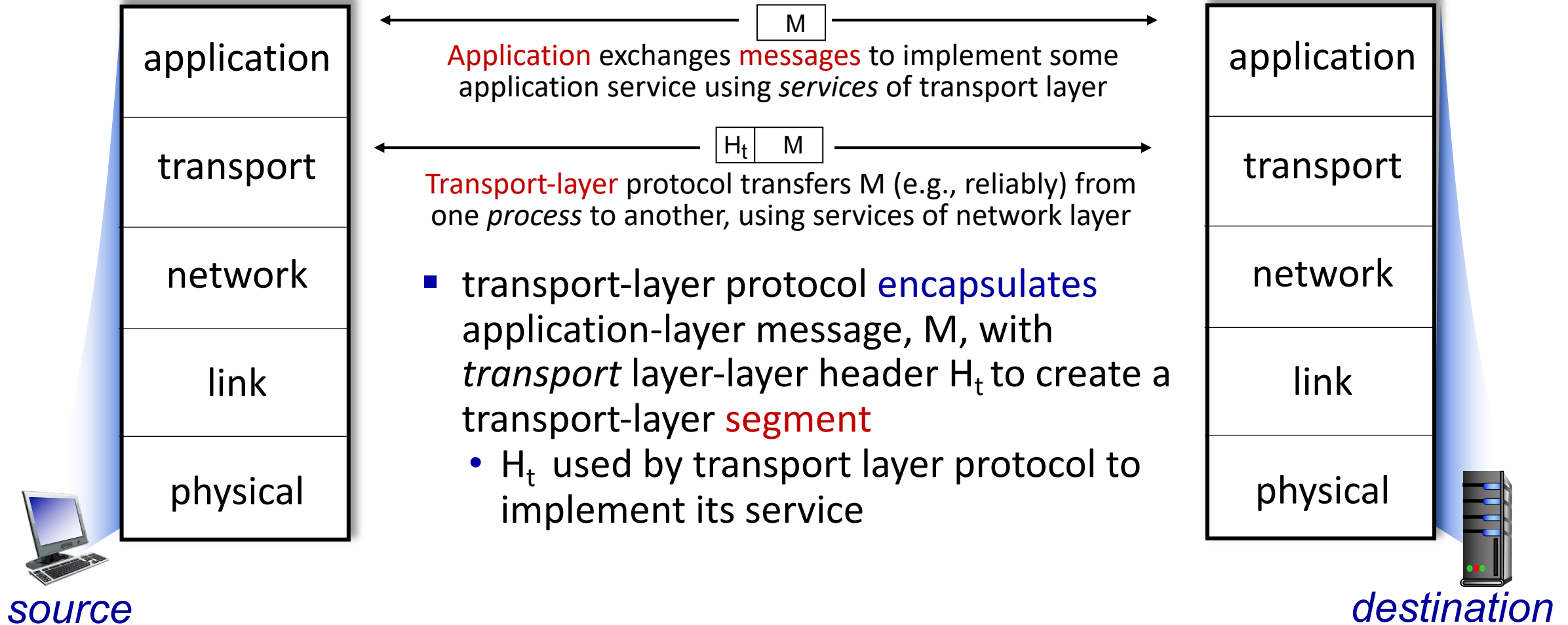
IEEE 802 standards “refine” the OSI data link layer.



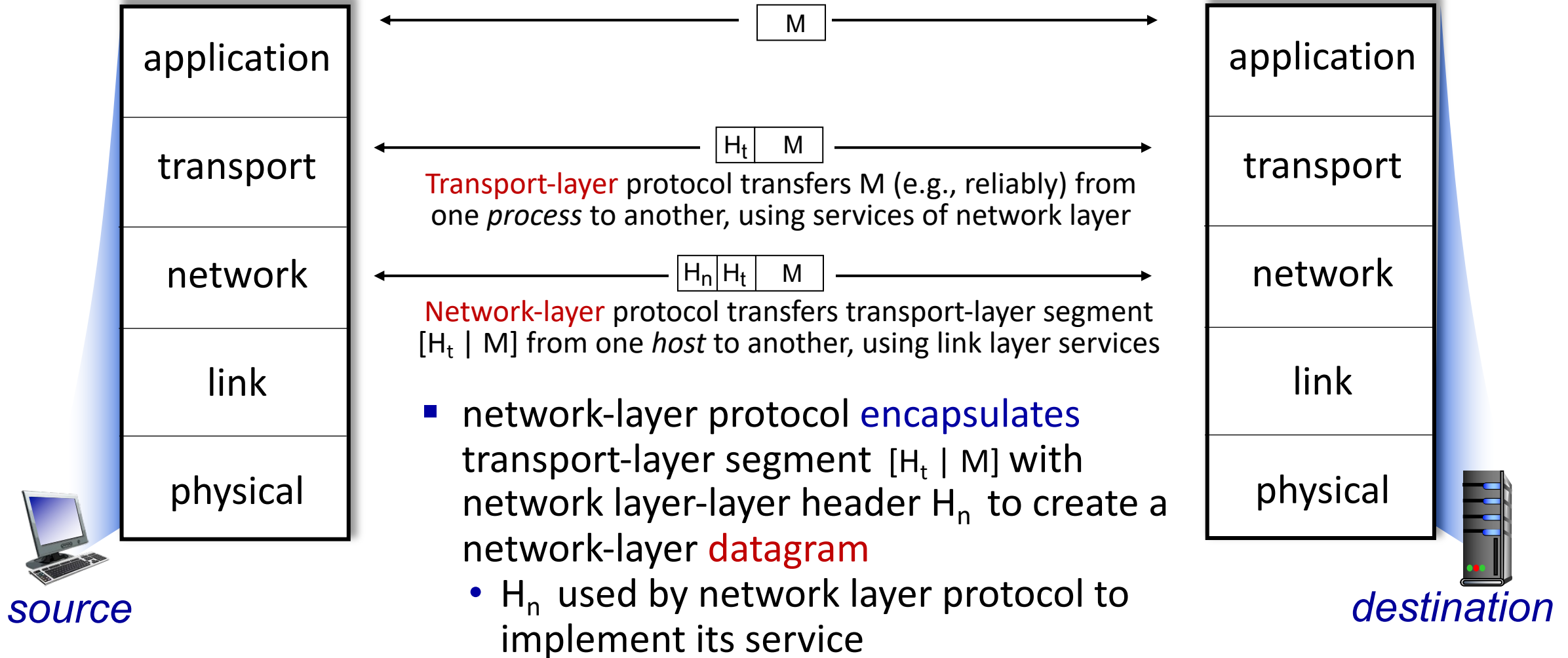
A TCP / IP / 802.3 Packet



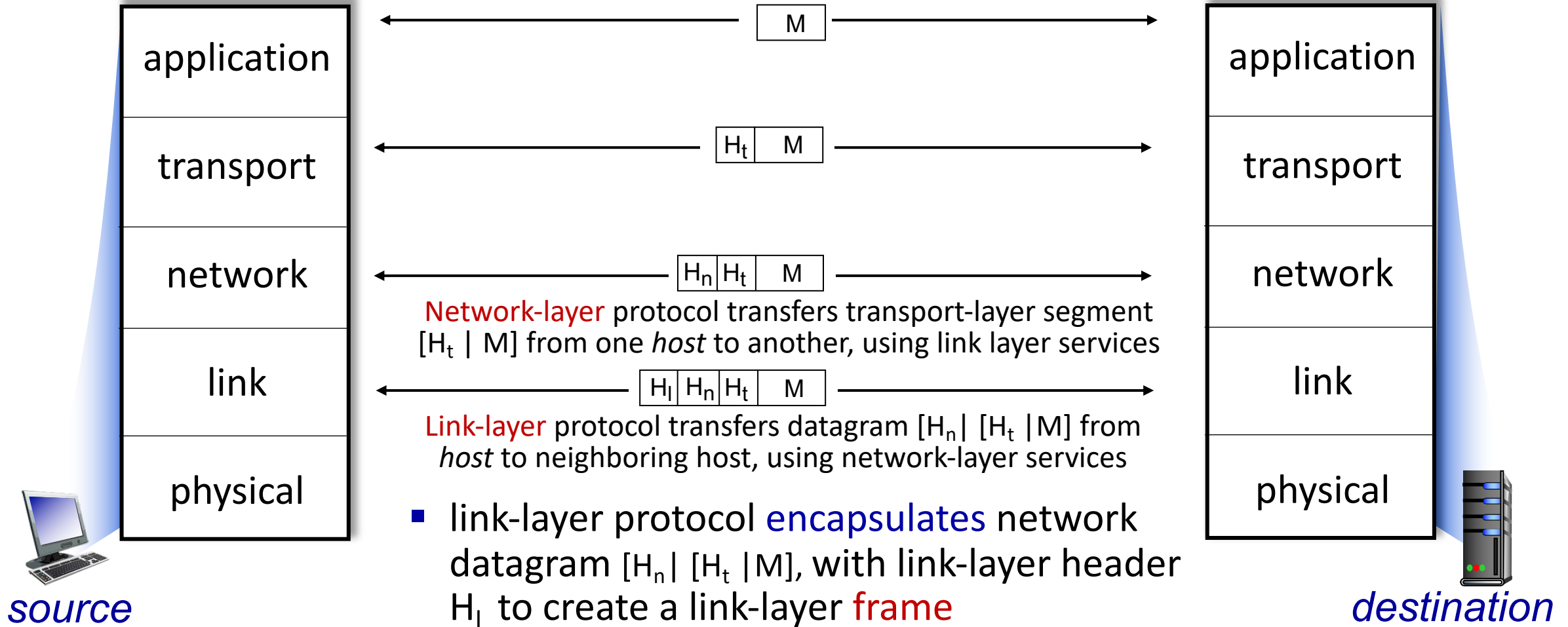
Services, Layering and Encapsulation



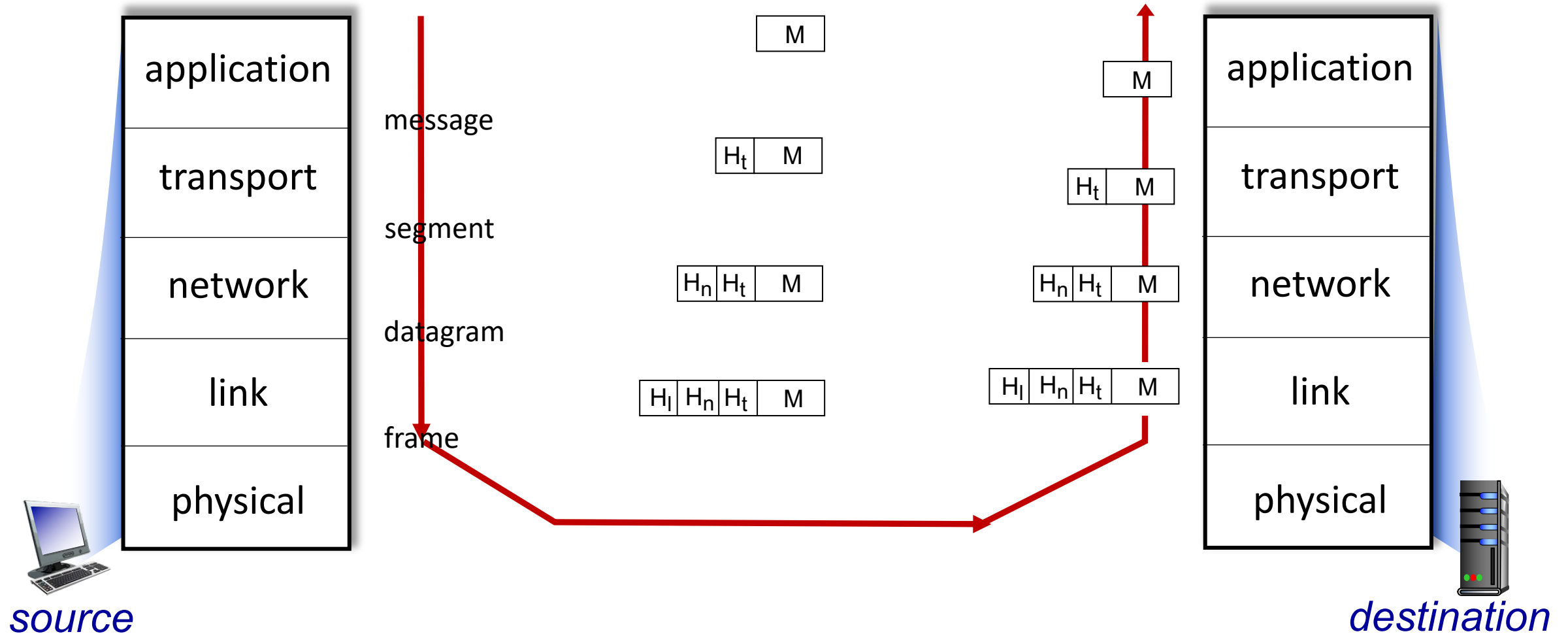
Services, Layering and Encapsulation



Services, Layering and Encapsulation

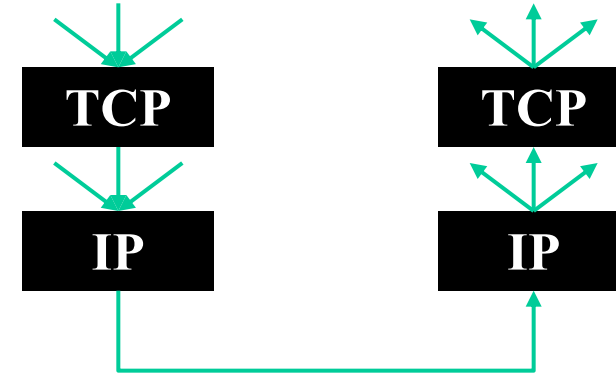


Services, Layering and Encapsulation



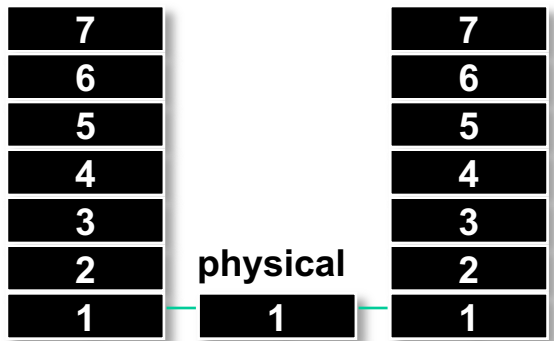
Multiplexing and Demultiplexing

- ❖ There may be multiple implementations of each layer.
 - How does the receiver know what version of a layer to use?
- ❖ Each header includes a demultiplexing field that is used to identify the next layer.
 - Filled in by the sender
 - Used by the receiver
- ❖ Multiplexing occurs at multiple layers. E.g., IP, TCP, ...

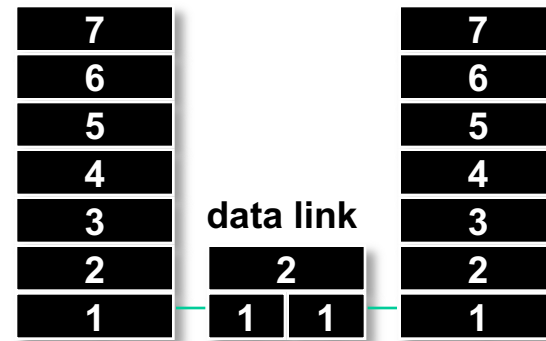


V/HL	TOS	Length
ID		Flags/Offset
TTL	Prot.	H. Checksum
Source IP address		
Destination IP address		
Options..		

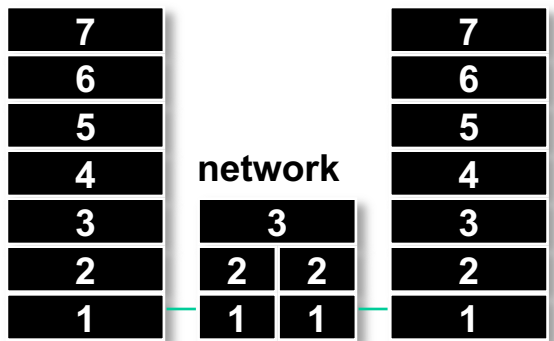
Internetworking Options



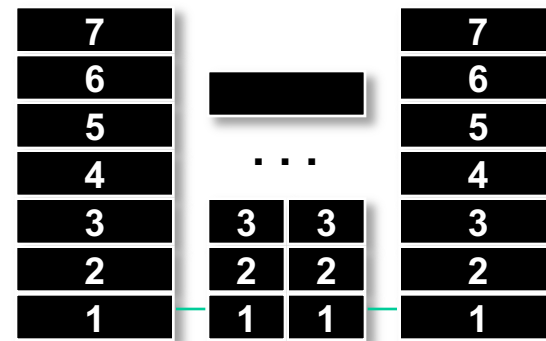
repeater / hub



bridge / switch
(e.g. 802 MAC)

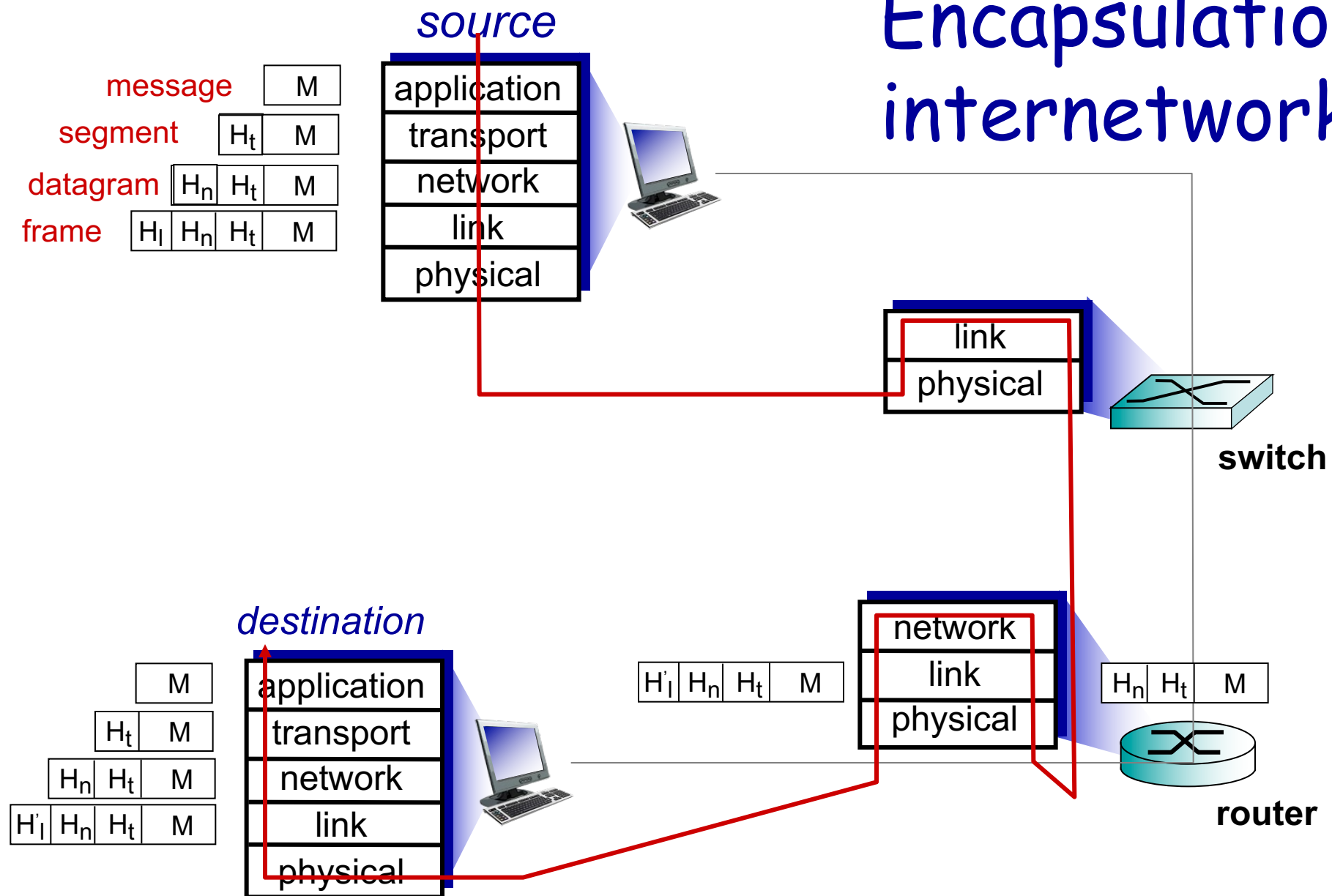


router

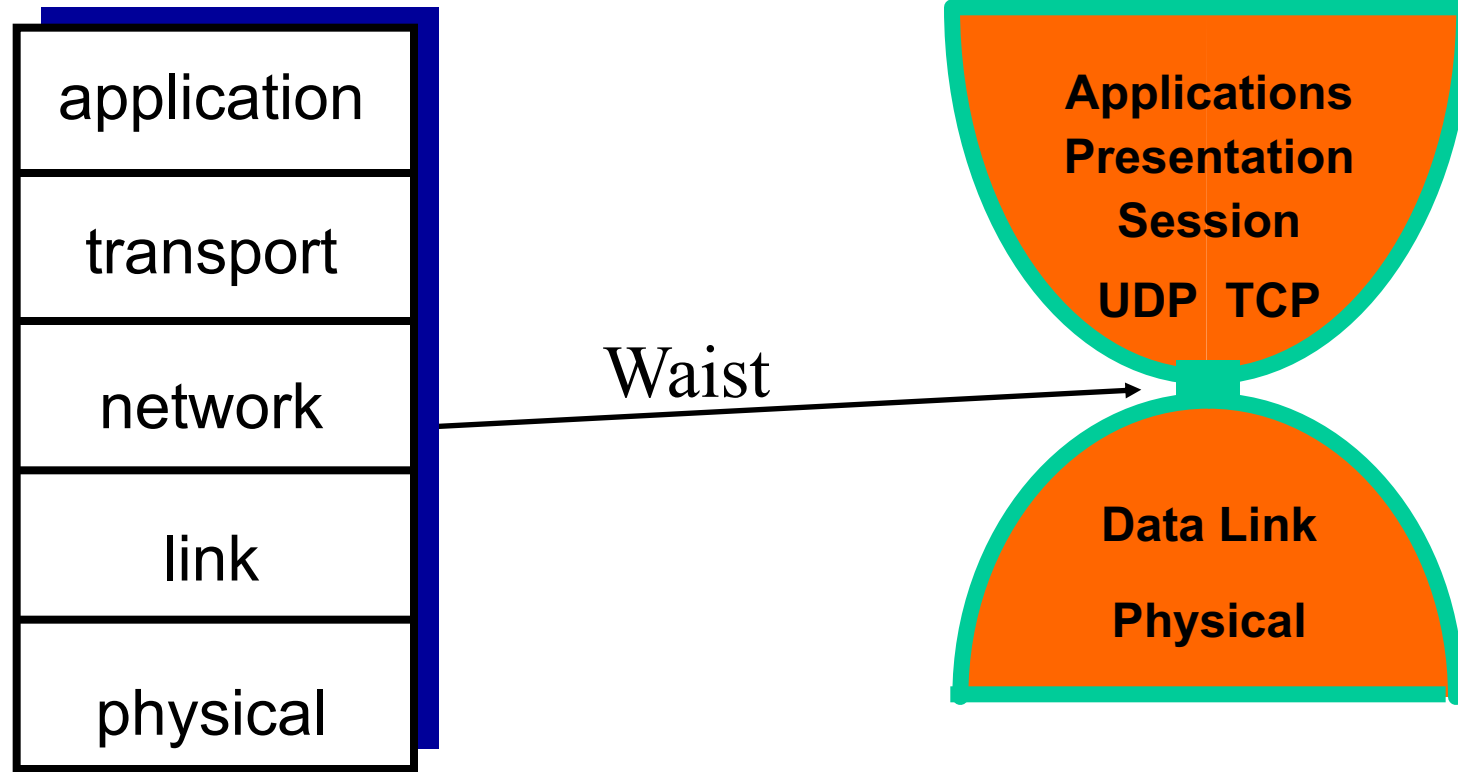


gateway

Encapsulation: internetworking



The Internet Protocol Suite



The waist facilitates Interoperability.

The Hourglass Model

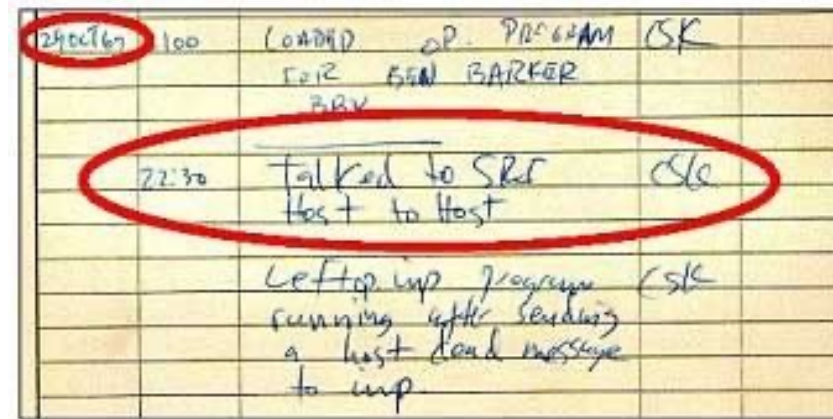
Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- **History**

Internet history

1962-1972: Early packet-switching principles

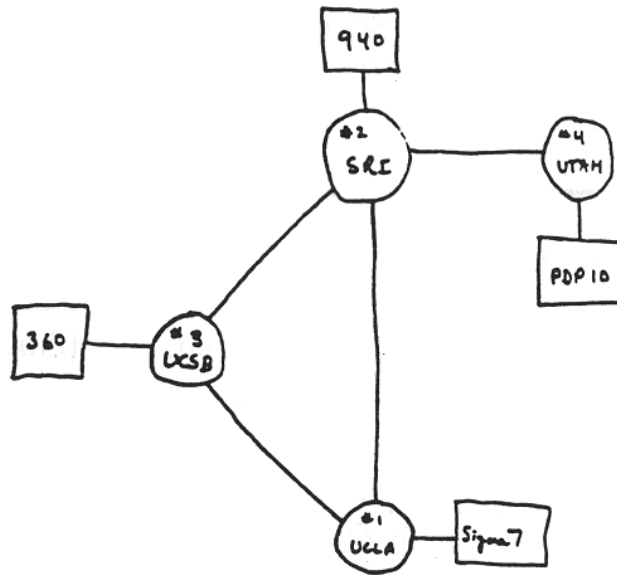
- ❖ **1961:** Kleinrock - queueing theory shows effectiveness of packet-switching
- ❖ **1964:** Baran - packet-switching in military nets
- ❖ **1967:** ARPAnet conceived by Advanced Research Projects Agency
- ❖ **1969:** first ARPAnet node operational
 - First link: UCLA-SRI



Internet history

1962-1972: Early packet-switching principles

- ❖ 1969: first ARPAnet node operational
 - Dec 5, 4 nodes



THE ARPA NETWORK

- ❖ 1972:
 - ARPAnet public demo
 - NCP (Network Control Protocol) first host-host protocol
 - first e-mail program
 - ARPAnet has 15 nodes

Internet history

1972-1980: Internetworking, new and proprietary nets

- ❖ 1970: ALOHAnet satellite network in Hawaii
- ❖ 1974: Cerf and Kahn - architecture for interconnecting networks



Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

**define today's Internet
architecture**

Internet history

1972-1980: Internetworking, new and proprietary nets

- ❖ 1976: Ethernet at Xerox PARC
- ❖ late 70' s: proprietary architectures: DECnet, SNA, XNA
- ❖ late 70' s: switching fixed length packets (ATM precursor)
- ❖ 1979: ARPAnet has 200 nodes
- ❖ 1983: deployment of TCP/IP



Bill Joy: TCP/IP in BSD UNIX

Internet history

1980-1990: new protocols, a proliferation of networks

- ❖ 1982: smtp e-mail protocol defined
- ❖ 1983: DNS defined for name-to-IP-address translation
- ❖ 1985: ftp protocol defined
- ❖ 1983 !
Jan 1, TCP/IP deployment
new national network
backbones: Csnnet, BITnet,
NSFnet, Minitel

❖ 1988:

- TCP congestion control
- First network worms
- 100,000 hosts connected to confederation of networks

Robert Tappan Morris



Internet history

1990, 2000 's: commercialization, the Web, new apps

- ❖ early 1990' s: ARPAnet decommissioned
- ❖ 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- ❖ early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960' s]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990' s: commercialization of the Web

late 1990' s - 2000' s:

- ❖ more killer apps: instant messaging, P2P file sharing (BitTorrent...)
- ❖ network security to forefront
- ❖ est. 50 million host, 100 million+ users
- ❖ backbone links running at Gbps

Internet history

2005-present: scale, SDN, mobility, cloud

- aggressive deployment of broadband home access (10-1000's Mbps)
- 2008: software-defined networking (SDN)
- service providers (Google, FB, Microsoft) create their own networks
 - bypass commercial Internet to connect "close" to end user, providing "instantaneous" access to social media, search, video content, ...
- enterprises run their services in "cloud" (e.g., Amazon Web Services, Microsoft Azure)
- rise of smartphones: more mobile than fixed devices on Internet. (In the second quarter of 2026, mobile devices (excluding tablets) accounted for 51.48% of global website traffic. [statista.com](https://www.statista.com))

Internet history

2005-present: scale, SDN, mobility, cloud

- ❖ short-range IoT devices, such as smart home assistants and wearable fitness trackers, were the most internet-connected devices worldwide, reaching 17.4 billion users. Mobile phones followed, with 8.62 billion connections. (2025, statista.com)
- ❖ more applications:
 - P2P applications: BiSkype (VoIP), PPLive (video)
 - Social networks: facebook, Twitter
 - YouTube, gaming,
- ❖ m2m traffic
- ❖ voice, video over IP
- ❖ wireless, mobility
 - increasing ubiquity of high-speed wireless access: 4G/5G, WiFi

Introduction: summary

covered a “ton” of material!

- ❖ Internet overview
- ❖ what's a protocol?
- ❖ network edge, core, access network
 - packet-switching versus circuit-switching
 - Internet structure
- ❖ performance: loss, delay, throughput
- ❖ layering, service models
- ❖ security
- ❖ history

you now have:

- ❖ context, overview, “feel” of networking
- ❖ more depth, detail *to follow!*